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WASTE AND BIOMASS MANAGEMENT & VALORIZATION

Layered double hydroxides and LDH-derived materials in chosen environmental applications: a review

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Abstract

With increasing global warming awareness, layered double hydroxides (LDHs), hydrotalcites, and their related materials are key components to reduce the environmental impact of human activities. Such materials can be synthesized quickly with high efficiency by using different synthesis processes. Moreover, their properties' tunability is appreciated in various industrial processes. Regarding physical and structural properties, such materials can be applied in environmental applications such as the adsorption of atmospheric and aqueous pollutants, hydrogen production, or the formation of 5-hydroxymethylfurfural (5-HMF). After the first part that was dedicated to the synthesis processes of hydrotalcites, the present review reports on specific environmental applications chosen as examples in various fields (green chemistry and depollution) that have gained increasing interest in the last decades, enlightening the links between structural properties, synthesis route, and application using lamellar materials.

Keywords Layered double hydroxides · LDH synthesis · Environmental applications · Adsorption · Hydrogen production

Introduction

Layered double hydroxides, also called LDHs or anionic clays, are particular lamellar materials made of octahedral sheets composed of metallic cations in the center of the octahedra. These materials can be found in nature and present the brucite ($Mg(OH)_2$) structure with divalent cations M^{II} (Mg^{2+} or Cu^{2+} for example) partly substituted by trivalent cations M^{III} (such as Al^{3+} or Fe^{3+}). This partial substitution induces a positive charge deficit that is compensated by anions, such as carbonates, located in the interlayer space along with water molecules. The easily accessible interlayer is one of the sources of the anion exchange capacity of the compounds and of its ability to capture and exchange anions. This characteristic is almost unique for inorganic materials. Indeed, clay minerals exhibit cation exchange capacity. Thanks to these characteristics, they are appreciated for their versatility, simplicity of tailoring, and low cost. The general formula of LDHs can be described as follows:

$$\left[M^{II}{}_{1-x}M^{III}{}_{x}\ (OH)_2\right]^{x+}\left[A^{m-}{}_{x/m}\cdot nH_2O\right]^{x-}$$

where M^{II} and M^{III} respectively represent the metallic divalent (Mg^{2+}, Cu^{2+}, Zn^{2+}) and trivalent (Al^{3+}, Fe^{3+}) cations, A the compensating anion, and x and m depend on the substitution rate between M^{II} and M^{III} cations.

Hydrotalcites are particular LDHs (characterized by a Mg:Al molar ratio of 3:1) in which magnesium is partly substituted by aluminum. Specific properties such as porosity and surface area can be tailored by varying the synthesis method and conditions, leading to a wide range of materials adapted to targeted applications. The structure can also be modified by changing the combination of the cations and the interlayer anions, in order to confer specific properties (Homsi et al. 2017; El Rouby et al. 2018; Sheng et al. 2019; Wu et al. 2019a).

LDHs can be synthesized by various techniques depending on the final requirements such as high crystallinity, low time and/or energy consuming reaction, and phase purity. The most common methods used to synthesize these materials are hydrolysis reaction, co-precipitation, hydrothermal synthesis, steam activation, microwave irradiation, and sol–gel. These fast and

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environmental-friendly synthesis processes, combined to high sorption capacities, allow the use of hydrotalcites and derived materials in various applications, such as adsorption of various molecules (organic and inorganic pollutants or dyes for example) or catalytic processes (also thanks to the presence of strong basic sites). In particular, these materials have been intensively studied as catalysts and catalyst supports based on their specific physicochemical features.

Mixed oxides, obtained from the calcination of the parent material, are most generally used in catalytic and environmental applications, due to the high temperatures required. The composition of the obtained oxides depends on the initial combination of M^{II} and M^{III} cations (El Rouby et al. 2018; Rahmanian et al. 2018; Sheng et al. 2019; Wu et al. 2019a), but also on the heating temperature and the targeted reaction (dehydration, dehydroxylation, removal of the interlayer anion, and oxide reformation). Considering the rising interest in using synthesized LDHs for various applications, this paper reviews the most common synthesis methods for their preparation and some chosen environmental applications (Fig. 1). Due to the wide panel of applications involving layered double hydroxides and LDH-derived materials, the list of applications here reported is not exhaustive. Moreover, certain applications have been already extensively reported in the literature and they could represent alone the subject of dedicated review articles (for example in the case of bio-diesel production; Helwani et al. 2009; Atadashi et al. 2013).

Synthesis techniques for hydrotalcite-like compounds

As previously explained, hydrotalcite (and more generally LDHs) can be of natural origin (alteration minerals) or synthetic. The main advantage in synthesizing LDHs is the ability to finely tune the mineral properties in view of a specific application. In this perspective, the appropriate combination of reactants has to be found to adapt the M^{II}/M^{III} molar ratio. The most important factors to be considered are the nature of the metallic cations, the pH, the temperature, the aging time, and the preparation method. Several synthesis methods have been developed during the last decades, but this review summarizes only those most commonly used and described in the literature.

Co-precipitation method

Co-precipitation is the most common method to synthesize LDHs. It consists in dissolving the inorganic salts in alkaline media, at constant or increasing pH. This reaction allows to control the morphology and particle size, depending on the supersaturation of the solution. Feitknecht reported for the first time in 1942 on the LDH synthesis by co-precipitation. [Mg-Al-CO_3] LDH was synthesized starting from diluted solutions of the reactants. The co-precipitation synthesis process has been studied years later by Gastuche et al. (1967) and Miyata (1975, 1980) by modifying parameters such as the concentrations of the reactants and the washing conditions. They also introduced the control of the pH as a key parameter for the formation of LDHs. Several studies have been performed later, focusing on the substitution of the type of metallic cations in the structure of the LDHs, as shown by Sato et al. (1988), in order to obtain several LDHs with different structural formulas ($M^{II} = Mg^{2+}$, Ni^{2+}, or Co^{2+} and $M^{III} = Al^{3+}$ or Fe^{3+}). Co-precipitation allows to finely tune the structure of the synthesized materials by controlling the M^{II}/M^{III} molar ratios, the type of interlayer anion, the synthesis duration, the temperature, and the pH of the solution (Thevenot et al. 1989; Ulibarri 2001; Zhao et al. 2003; Kloprogge et al. 2004; Sharma et al. 2007; Klemkaite et al. 2011).

Co-precipitation can be performed at low or high supersaturation conditions, depending on the desired crystallization state. In both cases, the initial concentration of the reactants has to be above the saturation state of the solution (supersaturation conditions). In any case, the pH of the solution must be controlled: a pH value too low does not allow the complete precipitation of all metallic ions, while a too high value leads to the leaching of one or more metallic ions. Basic media are preferred in most cases, but the optimal pH depends mainly on

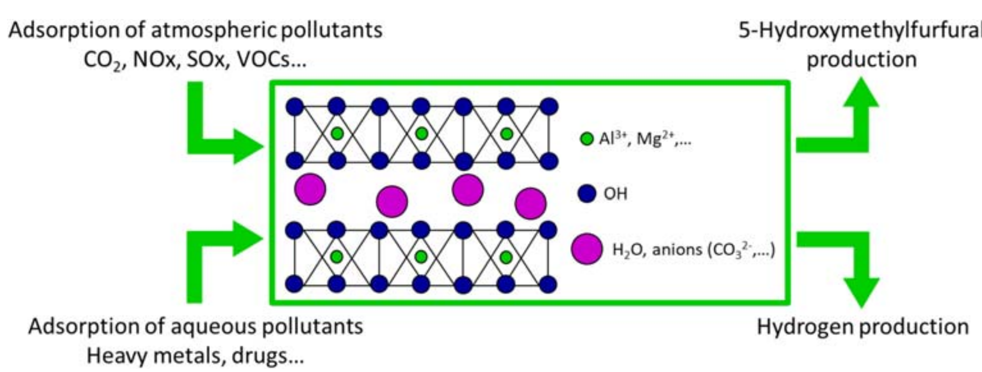


Fig. 1 Environmental applications of hydrotalcites discussed in this review

the metal cations used. Figure 2 presents the steps involved in the co-precipitation synthesis process. The co-precipitation method performed at low supersaturation conditions has been firstly documented in the 1990s by Kannan et al. (1995). This technique is performed by the slow addition of solutions containing the divalent and trivalent cations in the appropriate molar ratio, before adding an aqueous solution of the chosen interlayer anion (Kannan et al. 2004; Climent 2004; Kloprogge et al. 2004; Kovanda et al. 2005a; Perez-Lopez et al. 2006; Sharma et al. 2007; Wang et al. 2011; Klemkaite et al. 2011; Balsamo et al. 2012). An alkaline solution is then added in order to fix the pH during the reaction and to promote the co-precipitation of the two metal salts. This synthesis process has been well described by a schematic flowchart edited by El Rouby et al. (2018). The advantage of this method is the possibility to control the charge density; the pH needs to be kept constant under low supersaturation conditions by the addition of a mixture of NaOH and $NaHCO_3$ or Na_2CO_3 (Sharma et al. 2007; Wang et al. 2011; Klemkaite et al. 2011; Balsamo et al. 2012; Sheng et al. 2019; Li et al. 2019a). Materials obtained by co-precipitation at low supersaturation present a higher crystallinity than those obtained at high supersaturation conditions thanks to the continuous regulation of the pH of the synthesis solution. In the high supersaturation condition, the concentration of the solutions can be continuously increased. Another way is to add the solution containing the dissolved salts into a solution containing a small excess of alkali bicarbonates. This process leads to the formation of less crystalline materials than low supersaturation method, but gives rise to a high number of small aggregates due to the interlayer confinement effect of the carbonate ions (Zhang et al. 2008; Abelló et al. 2010; Solovov et al. 2018).

Co-precipitation synthesis can be considered an efficient method to obtain crystalline materials, particularly thanks to the long synthesis duration time (several days), the controlled synthesis pH, and the addition of posttreatments such as hydrothermal treatment for example (Kannan et al. 1995; Kloprogge et al. 2004). However, due to the large dimension of the obtained particles, the specific surface area values (between 50 and 100 $m^2\ g^{-1}$) are lower than those obtained by sol–gel synthesis (150–200 $m^2\ g^{-1}$) and discussed later in this review (Aramendía 2002; Park et al. 2018). The substitution with other metallic cations, such as Fe, Zn, Cu, and Ni, as demonstrated by Gevers et al. (2019), adds the possibility to modify the crystallite dimensions and influence the thermal stability of LDHs. Thermal treatments, such as hydrothermal treatment or microwave irradiation, can be performed to improve crystallinity and/or to reduce the synthesis time. Similar structures are then obtained even with shorter synthesis time (Climent 2004). This progress was reported by Tichit et al. (2002) by comparing microwave and hydrothermal treatments in co-precipitated hydrotalcites. The authors did not find any significant differences in the structural properties of the final material, despite the shorter synthesis time that represents a clear advantage. The thermal treatment presents a significant impact on the structural properties of the materials: Othman et al. (2006) compared the effect of the hydrothermal treatment on hydrotalcites prepared by co-precipitation and sol–gel methods. They showed that the specific surface area and pore sizes significantly increased in both cases, with a stronger impact on the samples synthesized by sol–gel.

Urea hydrolysis

Urea is a weak Brönsted base highly soluble in water. It can be used as a precipitating agent instead of sodium hydroxide to increase the pH through its thermal decomposition. The slow decomposition, starting around 90 °C, leads to an increase of the pH up to 10 (Costantino et al. 1998; Yang et al. 2004). Since urea decomposes slowly, it leads to a lower degree of supersaturation. Moreover, prolonged hydrolysis results in the formation of CO_3^{2-} anions in basic medium that act as interlayer anions. Therefore, urea enables an efficient pH control and the formation of mono-dispersed LDH materials with high crystallinity through a low-cost method (Adachi-Pagano et al. 2003; Yang et al. 2004; Zeng et al. 2009a; Xu et al. 2016; Park et al. 2018). The purity and crystallinity of hydrotalcite phases can also be optimized by tuning the M^{II}:M^{III} molar ratio, the urea concentration, and the aging time (Costantino et al. 1998; Zeng et al. 2009a; Berber et al. 2013; Montañez et al. 2014; Zhang et al. 2014; Bian et al. 2016; Xu et al. 2016; Park et al. 2018). As an example, Berber et al. (2013) identified the optimal synthesis conditions to obtain uniform and highly crystalline hydrotalcite particles by using a urea/metallic cation molar ratio of 2 and 12 h of

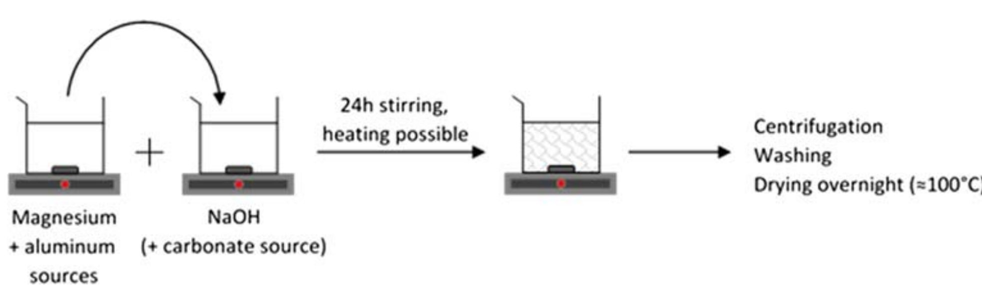


Fig. 2 Scheme of the co-precipitation method applied to LDH synthesis

aging time at 120 °C. Secondary phases have been observed with molar ratios lower or greater than 2, leading to lower crystallinity and specific surface areas of the materials. The main steps involved in hydrotalcite synthesis via urea hydrolysis are summarized in Fig. 3.

Most of the time, this technique is combined to others (in which urea is used only as a precipitating agent), such as co-precipitation or hydrothermal synthesis. The combination of urea hydrolysis and hydrothermal treatment has been well-documented by Berber et al. (2013) and Benito et al. (2006) in order to obtain uniform particles, but also by Rao et al. (2005) to find optimal Mg:Al molar ratios of 1:1 and 2:1 with uniform and small particle size. In the first article, microwave irradiation was used as heating source in the synthesis process, strongly shortening the synthesis time and the formation of pure and well-crystallized hydrotalcite phases presenting high specific surface areas. Further studies were performed by Montañez et al. (2014) by comparing the properties of mixed oxides obtained by calcination of a series of Ni–Al–Mg hydrotalcites synthesized either by co-precipitation or by urea hydrolysis. Enhanced redox properties, larger pore sizes, and smaller particles (interesting features in catalytic applications) were obtained on calcined samples prepared with urea hydrolysis. In general, the use of microwave irradiation allows to reduce the preparation time without impacting the structural and physical properties. This methodology is of great interest for industrial scaling-up (Yang et al. 2007; Benito et al. 2008). Hibino and Ohya (2009) combined urea hydrolysis with hydrothermal treatment in order to remove certain by-products (i.e., hydrated magnesium carbonate hydroxide phases). Further studies have been performed by Kloprogge et al. (2006) to show the evolution of the intercalated species depending on the hydrothermal treatment: $(NH_2)COO^-$ tends to directly form after precipitation, but slowly transforms into carbonated species during the hydrothermal treatment.

As a partial conclusion, urea hydrolysis is a slow reaction that leads to a low degree of supersaturation during the precipitation. It presents the advantage to provide large and thin platelets with a narrow particle size distribution in a shorter aging time than co-precipitation (Naseem et al. 2019). However, the decomposition of urea leads to the formation of CO_2 that reacts with water to form carbonates and acting as compensating anions (Thomas 2012). In this case, a further anion exchange may be required depending on the targeted application (Bish 1980; Iyi et al. 2004; Inayat et al. 2011).

Sol–gel method

The sol–gel method is appreciated for its low cost, low energy, and limited time consumption, for the possibility to obtain high purity materials, and for the possibility to tune their composition (Lopez et al. 1996; Paredes et al. 2006; Sharma et al. 2007; Ramos-Ramírez et al. 2009; Lee et al. 2016). This process allows to control the structural properties of the final materials by simply varying the chemical nature of the reactants and the aging time and by removing/adding reactant species. This technique consists in dissolving the desired metal sources (inorganic salts or metal organic compounds) in water, at room temperature (Fig. 4).

The concentration of the metal cations can be varied to tune the substitution ratio of M^{II} and M^{III} cations and to obtain a wide range of materials (Corma et al. 1994; Valente et al. 2007, 2009a; Sharma et al. 2007). The appropriate amount of acid or base is then added to the reactant solution during hydrolysis to favor the condensation reaction. The solution is finally aged during several hours or days, at room temperature or lightly heated up to 100 °C. Prince et al. (2009) proposed a general sol–gel method for the preparation of LDHs that can be adapted to obtain materials containing specific metallic cations and possessing a defined morphology in view of a specific application. Ramos-Ramírez et al. (2009) also evidenced the formation of small amounts of brucite phase, in addition to hydrotalcite, with a Mg/Al molar ratio of 2 and explained this behavior by the similar structure of the two materials. Synthetic hydrotalcites can also be calcined and rehydrated without losing their lamellar structure. During regeneration, they show a "memory effect," as evidenced by Climent et al. (2004) by using rehydrated sol–gel hydrotalcites for pharmaceutical applications.

Several parameters can be varied to improve the crystallinity (Kovanda et al. 2005b; Valente et al. 2009a) of the particles, to modify their morphology (Valente et al. 2007, 2009a, 2010),

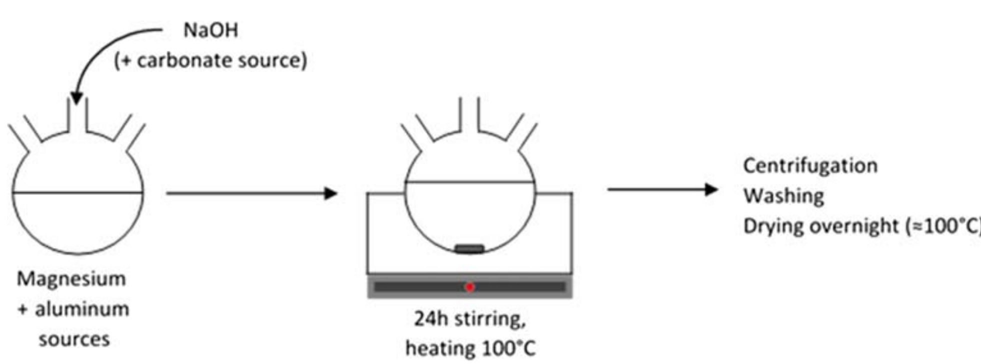


Fig. 3 Scheme of the synthesis of hydrotalcites via urea hydrolysis

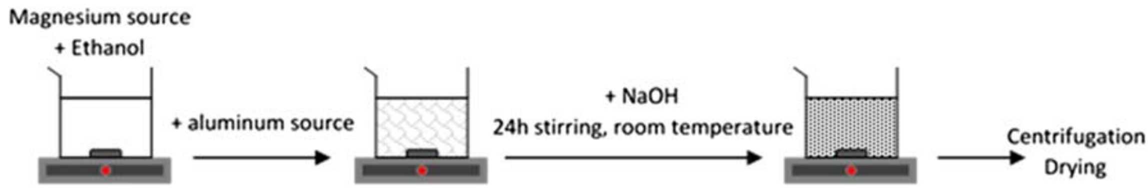


Fig. 4 Scheme of the synthesis of hydrotalcites by the sol–gel process

and to enhance the surface area (Corma et al. 1994; Lopez et al. 1996, 1997a; Climent et al. 2004; Paredes et al. 2006; Valente et al. 2007; Prince et al. 2009). Parameters, such as the composition of the aqueous media, the pH, and the aging time, were considered (Corma et al. 1994; Kovanda et al. 2005b). Lopez et al. (1997b) studied the impact of different alkoxides, such as magnesium ($Mg(OEt)_2$) and aluminum ($Al(NO_3)_3$), used as reactants, on the thermal stability of a series of hydrotalcites. Most of the time, the comparison between the synthesis techniques and the parameters evidences the interest of using one or the other specific method, generally sol–gel or co-precipitation (Jitianu et al. 2000; Prinetto et al. 2000; Bolognini et al. 2003; Othman et al. 2006; Valente et al. 2007, 2010; Smalenskaite et al. 2017). Smalenskaite et al. (2017) compared sol–gel and co-precipitation for the preparation of cerium-substituted Mg–Al LDHs; Othman et al. (2006) studied the impact of the thermal treatment on the structure of similar materials. In many cases, the sol–gel approach is preferred for its simplicity and the high quality of the materials obtained (Ramos et al. 1997; Othman et al. 2006; Valente et al. 2010; Lee et al. 2016; Smalenskaite et al. 2017).

The sol–gel synthesis method is appreciated for its high reproducibility, the high homogeneity and purity of the obtained compounds, the small size of the particles (nanoscale), and the high specific surface area (Prince et al. 2009; Smalenskaite et al. 2017; Valeikiene et al. 2019). However, lower crystallinity is generally observed through this synthesis route (Smalenskaite et al. 2017). Often and depending on the final application of the materials, additional steps such as sonication, microwave irradiation (Benito et al. 2006), or hydrothermal treatment are required.

Hydrothermal treatment

As previously reported, the synthesis of LDHs by co-precipitation, sol–gel method, or urea hydrolysis can be optimized by hydrothermal treatment in order to improve the crystallinity and the size of the crystallites (Kloprogge et al. 2004; Sharma et al. 2007). The treatment generally occurs under mild hydrothermal conditions, up to 200 °C, autogenous pressure, with duration between hours to several days. Figure 5 displays the hydrothermal synthesis process.

Besides the improvement of crystallinity, hydrothermal synthesis can be directly used as a synthesis method, in order to form LDH phases with increased particle sizes. It consists in dissolving the metallic salts into a solvent (generally water) before putting the solution into a stainless-steel reactor and heating under hydrothermal conditions, generally under medium temperatures (30–300 °C) and steam pressure, with different synthesis times (few hours to several days) (Xu and Lu 2005; Liao et al. 2012; Bontchev et al. 2003; Bankauskaite and Baltakys 2011; Lin et al. 2019; Budhysutanto et al. 2010; Roelofs et al. 2001; Jang et al. 2014). Several reactants such as inorganic salts and minerals can be used as metal sources. Most commonly, metal oxides and hydroxides (Xu and Lu 2005; Budhysutanto et al. 2010; Jang et al. 2014; Labuschagné et al. 2018) or nitrate salts (Roelofs et al. 2001; Bontchev et al. 2003; Lin et al. 2019) are used as reactants, but other sources can be used: Ogawa and Asai (2000) used for example natural minerals (brucite and gibbsite) as starting materials to intercalate organic guests. Liao et al. (2012) also used natural brucite and $Al(OH)_3$ as synthesis reactants to study the morphology and structural properties of hydrotalcites obtained under different synthesis conditions.

Like for the other synthesis techniques, the Mg:Al molar ratio and the interlayer anions can be modified in order to obtain a wide range of hydrotalcites and LDHs with different structural properties. Bontchev et al. (2003) studied the impact of several interlayer anions (Cl^-, Br^-, I^-, NO_3^-) on the physicochemical characteristics of hydrotalcites. Moreover, this study showed the possibility to mix two different types of compensating anions to finely tune the properties of these compounds. The hydrothermal synthesis of hydrotalcites with different soluble and insoluble magnesium and aluminum reactants in water has been studied by Bankauskaite and Baltakys (2011), with a fixed Mg:Al molar ratio of 3:1. The nonformation of hydrotalcite phases was observed in the presence of certain reactants. It was demonstrated that alumina and magnesium aluminum hydroxide were preferentially formed when using $4MgCO_3{\cdot}Mg(OH)_2{\cdot}5H_2O$, even after 24 h of hydrothermal treatment at 200 °C. The choice of the appropriate metal salts is the most important parameter to be considered. More recently, the impact of the synthesis temperature (from 35 to 140 °C) has been tested by Lin et al. (2019) to synthesize CuZnAl hydrotalcite-like catalysts for arsine abatement. The authors concluded on a significant influence of the synthesis conditions on the arsine removal efficiency without a clear explanation. The problem comes from the multiplicity of the parameters varied during the synthesis of the catalysts

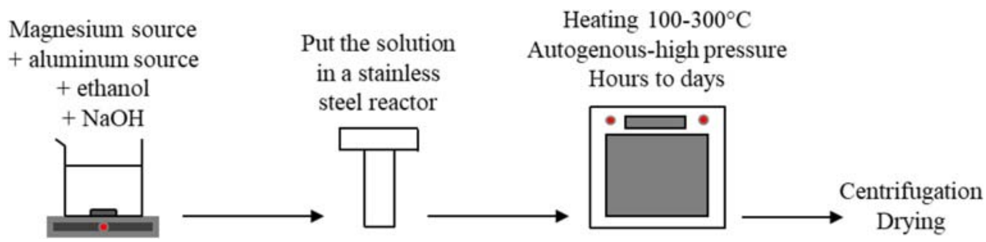


Fig. 5 Hydrothermal synthesis process applied to hydrotalcite synthesis

(temperature, calcination, time, molar ratios), making the results difficult to interpret. The most recent studies focus on the green synthesis of hydrotalcites and derived materials for environmental concerns, as demonstrated by Labuschagné et al. (2018) by using untreated magnesium oxide and aluminum hydroxide for hydrothermal dissolution/precipitation in water.

The advantages of using this synthesis process as additional treatment are the higher LDH crystallinity, crystallite dimension, and purity of the samples (Kovanda et al. 2005b; Sharma et al. 2007). In general, a higher content of hydrotalcite is observed by increasing the preparation temperature and the time of the hydrothermal treatment (Kovanda et al. 2005b). Unfortunately, the energy required to heat up the solution and the long synthesis time (from hours to days) represent drawbacks for the industrial implementation of the hydrothermal treatment.

The recent advancements and progresses on the synthesis methodologies have been reviewed in this section. Moreover, each synthesis method has been detailed in order to enlighten the advantages and drawbacks in terms of structural and textural properties. The modulation of these features is crucial to drive the performances toward specific environmental and industrial applications. Table 1 lists the positive and negative effects of the various synthesis routes on the structural and textural properties of LDHs.

Use of hydrotalcites and derived materials in environmental applications

The depletion of fossil fuels and their environmental impact (global warming, pollution, health concerns…) are currently boosting the studies on alternative solutions and clean synthetic routes for the production of nonfossil carbon sources. Recent international conventions agreed to the Kyoto Protocol and intended to reduce by 2050 the global emissions up to 50–60% (of those measured in 2006). The necessity in finding alternative environmental applications based on easy-to-use materials becomes more than urgent. Several reactions can be performed on layered materials, but their low thermal stability makes their implementation difficult. Clay minerals are generally stable up to 200 °C due to the loss of the interlayer water and the consequent irreversible degradation of the structure at temperatures higher than 200 °C. The presence of specific metal elements in the structure of the LDHs can enhance the catalytic properties, depending on the applications, as mentioned by Xu et al. (2011). Moreover, due to their composition (mainly metallic cations), LDHs can be calcined to form mixed oxides presenting enhanced catalytic properties. Even if much more environmental-related processes can be carried out in the presence of lamellar materials, only a few of them are discussed in this review. Some specific environmental applications covering various technological fields have been chosen to be reported in the present review article such as

Table 1 Main structural and textural impacts of different synthesis processes on LDHs

Synthesis method	Advantages	Drawbacks
Co-precipitation	High crystallinity with posttreatment	Large particles (formation of aggregates)
	Easy insertion of metallic elements	Low specific surface area
Use of urea	Large thin platelets	Formation of CO_2 by urea decomposition, formation of carbonates as counteranions
	Narrow particle size distribution	
Sol–gel	High homogeneity	Low crystallinity
	High purity	Additional treatments required (microwave irradiation, ultrasonication, or hydrothermal treatment)
	Small particles (nanoscale) = high specific surface area	
	Short times (hours)	
Hydrothermal	Increases crystallinity	High energy (heating) and time (hours to days) demand
	Increases particle size	More efficient as an additional treatment of other synthesis methods
	Increases purity	

the adsorption/decomposition of gaseous pollutants (CO_2, NO_x, and SO_x adsorption, and elimination of VOCs), the adsorption of pollutants in aqueous media (as the adsorption of heavy metals and dyes), and the production of green energy vectors and/or chemical building blocks (hydrogen production and formation of 5-hydroxymethylfurfural).

Adsorption of atmospheric pollutants

CO_2 adsorption—capture and storage

Among the possible uses of hydrotalcite-like compounds and their derived mixed oxides, CO_2 adsorption gains interest. Methods to safely control and dispose of carbon dioxide emissions have been widely studied in the last few years, mainly due to the increase of consciousness in climate changes (Wang et al. 2017). Among all the emitted greenhouse gases, the contribution of carbon dioxide to global warming has been estimated to be more than 60%. For example, steam reforming of hydrocarbons (Marquevich et al. 2001; Ashok et al. 2008; Li et al. 2011), that is the most suitable process for hydrogen production, releases high amounts of carbon dioxide. Considering the large field of materials tested for CO_2 capture and storage, zeolites, hydrotalcite, and LDH-derived mixed oxides are widely studied thanks to their high surface area, pore structure, and charge density (Mao and Tamaura 1993; Tsuji et al. 1993; Yong et al. 2002; Wang et al. 2010; Dantas et al. 2015; Yang et al. 2019). Several reviews on CO_2 adsorption explain that an appropriate adsorbent should satisfy several criteria such as (1) low-cost materials, (2) fast kinetics, (3) high adsorption capacity and selectivity, and (4) thermal and chemical stability toward several cycles (Yong et al. 2002; León et al. 2010; Wang et al. 2017; Yang et al. 2019). The adsorption of CO_2 requires materials with large specific surface area, but also a high number of accessible basic sites. Regarding the synthesis method adapted to this application, a high number of active sites and large particles are required for better adsorption efficiency. Thus, the co-precipitation method represents the best solution to reach these requirements, especially when the LDH structure is modified by the addition of metal cations. The higher the energy of active sites, the stronger CO_2 interacts with the sites and remains trapped. Recently, zeolites have been reported as efficient CO_2 adsorbents thanks to their size, charge density, and tunable chemical composition (Jiang et al. 2018; Yu et al. 2018). Hydrotalcite-based materials are thereby interesting alternative materials to zeolites, as their chemical composition can also be finely tuned. CO_2 adsorption performed on these materials is a natural phenomenon, as hydrotalcite can adsorb carbon dioxide thanks to the reaction of atmospheric CO_2 with the compensating ions present in the interlayer space. This has been evidenced by Ishihara et al. (2013) by intercalating ^{13}C carbonate anions in the hydrotalcite structure. They were exchanged with carbonate anions derived from atmospheric CO_2 within 1 day, showing the dynamism of the carbon cycle in nature. Moreover, these materials can be regenerated several times, without losing their adsorption capacity and selectivity toward CO_2, are hydrothermally stable, and lead to fast sorption kinetics. A huge amount of studies has been performed on these materials in order to find the most efficient preparation route for CO_2 capture, such as impregnation of commercial hydrotalcites (Bhatta et al. 2015), synthesis of tunable Mg/Al hydrotalcites (Tsuji et al. 1993; Moreira et al. 2006; Dantas et al. 2015), and mixed oxides derived from hydrotalcites (León et al. 2010; Gao et al. 2013; Radha and Navrotsky 2014; Colonna et al. 2018). Table 2 summarizes the synthesis conditions, the specific surface area, and the CO_2 adsorption capacity of the LDHs and related mixed oxides. It shows that the surface area is strongly linked to CO_2 adsorption capacity: a higher surface area is related to an increase in the sorption capacity.

The preliminary studies of CO_2 adsorption by hydrotalcite, layered double hydroxides, and related materials have been performed by Mao and Tamaura (1993) by varying the M(II)/M(III) molar ratio. Consequently, the layer charge of the synthesized hydrotalcites was modified. The tuning of the composition resulted in a wide range of materials capable of chemically adsorbing CO_2 with different adsorption capacities (from 0.4 to 1.5 mmol g^{-1}). On the same year, Tsuji et al. (1993) synthesized various LDHs by changing the M(II) cation nature. The selectivity toward carbon dioxide adsorption depended on the cation in the following order: Cu–Al ~ Zn–Al < Co–Al < Mg–Al < Ni–Al, and was related to the thermal behavior of the various compounds. Moreover, the sorption capacities of these materials have been drastically increased by removing the carbonate counterions in the Mg–Al and Co–Al LDHs. Several years later, Wang et al. (2010) studied the substitution of divalent metal cations M(II) by trivalent metal cations M(III) (Al^{3+}, Ga^{3+}, Fe^{3+}, and Mn^{3+}) in CO_2 adsorption at high temperature, performed on calcined samples.

A few years after Tsuji et al. (1993), studies about CO_2 adsorption were mainly focused on industrial applications and enlightened the influence of structural parameters derived from the thermal treatment of LDHs, leading to the formation of related mixed oxides. Ram Reddy et al. (2006) showed that, depending on the applied treatment temperature, the samples exhibited different sorption capacities and reversibility. The highest sorption capacities were observed on samples pretreated at 400 °C and a sorption temperature of 200 °C, leading to CO_2 adsorption of 0.486 mmol g^{-1}. The pretreatment conditions showed also an impact on the regeneration behavior (cycling studies). Two years later, the same team (Ram Reddy et al. 2008) studied high-temperature CO_2 adsorption in industrial conditions. An increase of the sorption efficiency was observed, from 0.61 to 0.71 mmol g^{-1}, in the presence of dry and wet CO_2 respectively, with more than

Table 2 Synthesis parameters of the LDHs and mixed oxides cited in the section "CO_2 adsorption—capture and storage"

Reference	Type of material	Surface area ($m^2\ g^{-1}$)	Adsorption capacity ($mmol\ g^{-1}$)	Synthesis method/structural modifications
Mao and Tamaura (1993)	MgAlFe LDH	–	–	Co-precipitation at various Al/(Mg+Al) molar ratios and pH 10
Tsuji et al. (1993)	Various LDHs	–	–	Co-precipitation with change of the M(II) cations: Mg^{2+}, Ni^{2+}, Zn^{2+}, Co^{2+} at pH 10
Wang et al. (2010)	Mg-M(III) LDHs	82.4–148.6	0.050–0.462	Co-precipitation with change of the M(III) cations: Al^{3+}, Ga^{3+}, Fe^{3+}, Mn^{3+}, Cr^{3+}, Ce^{3+}, La^{3+} at constant pH 10
Ram Reddy et al. (2006)	Calcined MgAl LDHs	63–167	0.231–0.486	Co-precipitation and calcination at different temperatures from 200 to 600 °C for 4 h
Ram Reddy et al. (2008)	Calcined MgAl LDHs	–	–	Co-precipitation and calcination at 400 °C for 4 h
Dadwhal et al. (2008)	Calcined MgAl LDHs	–	0.7	Co-precipitation and calcination at 500 °C for 4 h
León et al. (2010)	Calcined MgAl LDHs	62–190	0.4–1.15	Co-precipitation at low (pH 10) and high supersaturation and calcination at 450 or 700 °C for 7 h. Anion exchange by K^+ or Na^+ prior to calcination
Gao et al. (2013)	Calcined MgAl LDHs	154–239	0.30–0.72	Co-precipitation followed by hydrothermal treatment at 120 °C for 6 h, urea method, or urea decomposition followed by microwave irradiation at 120 °C 200 W for 30 min
Ramírez-Moreno et al. (2014)	MgAl LDHs	136.6–296.3	0.2–0.75	Co-precipitation at low supersaturation and at pH 9.5
Bhatta et al. (2015)	MgAl LDH supported in coal-derived graphitic material	41–219.6	0.48–1.10	Co-precipitation at low supersaturation and at pH 10–12
Dantas et al. (2015)	MgAl LDHs expanded with a polymer	50.1–61.1	0.72–1.36	Co-precipitation at low supersaturation
Tang et al. (2018)	MgAl LDHs enlarged by SDS ions before grafting 3-APS	–	1.55–2.09	Co-precipitation at low supersaturation and at pH 10
Thouchprasitchaia et al. (2018)	TEPA-functionalized calcined MgAl LDHs (hydrotalcite)	5.7–165.2	2.03–6.03	Co-precipitation at low supersaturation
Wang et al. (2012a, 2012b)	Amine-modified MgAl LDHs via exfoliation or anionic surfactant-mediated routes	–	0.75–1.8	Co-precipitation at low supersaturation and at pH 10.3
Wu et al. (2013)	K-promoted hydrotalcite	–	1.15	Commercial hydrotalcite

90% of the original capacity recovered by regeneration. Dadwhal et al. (2008) reported on the influence of particle size on CO_2 adsorption and analyzed the formation of large particles by co-precipitation. Mixed oxides generally exhibit higher sorption capacities due to the higher specific surface area than the fresh samples, but the analysis conditions (sorption temperature, mixture of gas, etc.) affect the CO_2 adsorption capacities in the same way as on noncalcined samples, as demonstrated by León et al. (2010). The obtained capacities, in the 0.58–1.15 mmol g^{-1} range at 50 °C and 0.40–0.84 mmol g^{-1} range at 100 °C, depend on the pretreatment temperature, 450 or 700 °C, respectively. Regarding the reversibility of these materials, the co-precipitation method at high supersaturation conditions is more adapted than that at low supersaturation, due to the formation of weaker basic sites for the latter. As previously explained, Wang et al. (2010) studied the influence of M(III) cation type toward high-temperature CO_2 adsorption on calcined samples at different temperatures, from 250 to 400 °C. Different phases have been observed depending on the calcination temperature: $MgFe_2O_4$ and $MgMnO_4$ with coexistence of MgO with the appropriate M(III) cations, whereas only MgO phases have been observed in MgAl and MgGa LDHs. Moreover, the M(III) cation nature did not affect the sorption capacity (around 0.4 mmol g^{-1}), except for the calcined MgGa LDH that gave lower values of CO_2 sorption capacity due to its lower thermal stability. Gao et al. (2013) discovered the optimal parameters to obtain hydrotalcites with improved CO_2 adsorption capacity: synthesis by the co-precipitation method, at high supersaturation conditions, with a Mg/Al molar ratio between 3 and 3.5, and a pre-calcination temperature of 400 °C (The CO_2 adsorption measurements were performed during 5 h at 200 °C). The operating conditions (temperature and pressure) impact the adsorption capacity of CO_2, too: Ramírez-Moreno et al. (2014) stated that high pressures (up to 4350 kPa) and temperatures (30–350 °C) are related to higher CO_2 adsorption levels, with a maximum value of 5.76 mmol g^{-1} measured at 300 °C and 3450 kPa. Higher values can be reached at higher temperatures and pressures and are related to structural changes in the LDH framework (formation of magnesium oxides and periclase for example).

The most recent research focuses on modifications of the structure of LDHs synthesized by co-precipitation in order to increase their CO_2 adsorption capacity. Similar to simple ion-exchange reaction, the potassium impregnation consists in putting the synthesized materials into a solution with a defined concentration of potassium salt during a few hours, followed by the washing of the salt excess. K_2CO_3 impregnation seems to increase the number of surface active sites, despite a decrease in the specific surface area. This impregnation leads to high adsorption capacities, as demonstrated by Bhatta et al. (2015): values up to 1.10 mmol g^{-1} at 300 °C for the initial sorption cycle and of 0.42 mmol g^{-1} for the following 9 cycles of adsorption/desorption were measured. Polymer addition can modify the textural properties of layered materials and contribute to the enhancement of the CO_2 sorption capacity, as shown by Dantas et al. (2015) by the addition of a copolymer that increased the CO_2 uptake from 0.72 to 1.36 mmol g^{-1}. Tang et al. (2018) increased the basal spacing of LDHs by exchanging anions with dodecylsulfonate ions in order to chemically graft (3-aminopropyl)triethoxysilane (APS) and enhance the CO_2 sorption capacity (up to 2.09 mmol g^{-1} at 50 °C). The presence of amino groups in the interlamellar space strongly contributed to the capture of carbon dioxide through the zwitterion mechanism and weak bonding. Thouchprasitchaia et al. (2018) obtained a series of calcined teraethylenepentamine (TEPA)-supported hydrotalcites by optimizing the synthesis parameters. At first, increasing the TEPA loading led to higher adsorption capacities, explained by the availability of more basic sites with a higher affinity for CO_2. A too high TEPA content resulted in the decrease of the sorption capacity due to the steric hindrance of TEPA that blocks the access of CO_2 to the sorption sites.

Among the large panel of industrial applications, sorption-enhanced reaction processes (SERP) gained huge interest in the last decade, especially thanks to novel modifications introduced into LDHs' structure. SERPs have been developed for the efficient conversion of methane to hydrogen. The main drawback of this process is the formation and release of large amounts of CO_2, then materials with high carbon dioxide sorption capacity are crucial for this application. The first amine modifications of LDHs for CO_2 adsorption performance toward SERP applications have been performed by Wang et al. (2012a, 2012b) via two different routes: anionic surfactant-mediated synthesis and exfoliation. In the first case, low CO_2 adsorption capacities (up to 0.6 mmol g^{-1}) have been recorded, mainly due to the presence of sodium dodecyl sulfate (SDS) in the interlayer space. SDS can be then removed using monoethylamine extraction (ion-exchange reaction), leading to higher sorption values (between 1.15 and 1.4 mmol g^{-1}) (Fig. 6). On the opposite, the exfoliation route in the presence of toluene allows to graft amine groups on the surface of LDHs giving high sorption capacities toward CO_2 (around 1.7 mmol g^{-1} at 80 °C). The two studies underline the importance of the interlayer space in LDHs: sodium dodecyl sulfate and aminosilane molecules, present in this area, cover the main part of the basic sites involved in the CO_2 adsorption. Consequently, the sorption capacity is decreased. It is important to consider the steric hindrance when considering lamellar materials. Most of the time, K-promoted hydrotalcites are interesting adsorbents in SERP, as the potassium presence leads to sorption capacities higher than 1 mmol g^{-1}, as demonstrated by Wu et al. (2013). Moreover, the adsorption capacity and kinetics were not affected even after 10 cycles, which is interesting for SERP applications. Coenen et al. (2017a, b) also published about K-promoted hydrotalcites in a series of publications for sorption-enhanced water–gas shift. This article points out the importance of chemisorbed CO_2 and H_2O, strongly linked to the specific adsorption sites. The presence of four different adsorption site types was identified: two site types with high affinity for CO_2, one for H_2O, and one on which CO_2 and H_2O can competitively adsorb. The CO_2 adsorption sites can be easily regenerated in the presence of N_2, and the cycling is favored at high temperatures. In the most recent study of the same group (Coenen et al. 2018), the adsorption processes were studied by infrared spectroscopy, showing once again the presence of the four types of adsorption sites. The presence of water molecules enables the decomposition of the strong bonds with carbonate sites, ameliorating the cycling behavior. The comparison between LDHs and zeolites has been recently performed by Megías-Sayago et al. (2019), and the importance of the surface area was evidenced. While zeolites exhibited surface areas around 360 m^2 g^{-1} and adsorb CO_2 up to 1.142 mmol g^{-1}, MgAl and CaAl LDHs presented lower specific surface areas (50 and 21 m^2 g^{-1}) and adsorb carbon dioxide up to 0.68 and 0.10 mmol.g^{-1}, respectively.

Table 3 and Fig. 7 display the CO_2 adsorption capacities of various LDHs reported in the literature. Nonmodified materials exhibit values between 0.5 and 1 mmol g^{-1}, while modified LDHs present much higher values, up to 6.1 mmol g^{-1}.

As demonstrated by various studies, LDHs are interesting alternatives to zeolites and activated carbons for CO_2 adsorption and storage. Adsorption capacities between 0.5 and 6 mmol g^{-1} could be measured depending on the structural modifications, the pretreatments performed prior to adsorption, and the configuration of the adsorption experiments

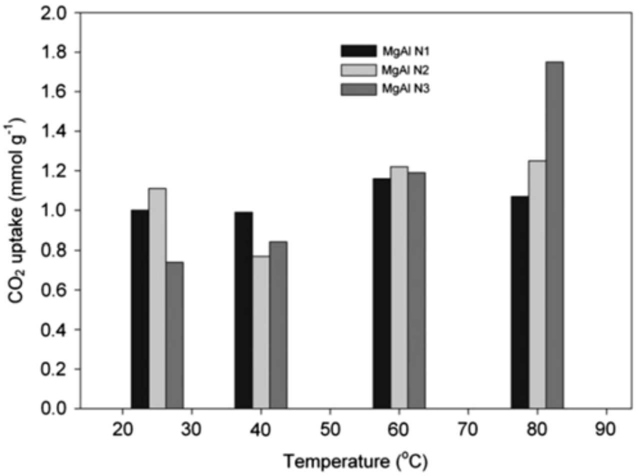


Fig. 6 CO_2 adsorption by amine-modified MgAl hydrotalcites at different temperatures (from Wang et al. 2012b)

(operating temperature and pressure, gas composition, time of adsorption). Older papers stated on the optimal synthesis conditions and co-precipitation performed at high supersaturation seems to be the most adapted method. More recent research focuses on sorption-enhanced reaction processes, due to the need to minimize the release of CO_2 while producing hydrogen from methane. Regarding the synthesis process, co-precipitation allows to obtain large and crystalline particles, increasing the amount of basic sites that are required to reach large amounts of CO_2 adsorbed. The calcination of such materials leads to the formation of mixed oxides with higher surface areas and an increased number of basic sites. One of the most important parameters to take into account in this application is the specific surface area that is strongly related to the CO_2 adsorption capacity. Moreover, Table 2 evidences slightly higher adsorption capacities for the materials synthesized by co-precipitation at high supersaturation conditions than for those prepared at low supersaturation conditions, except in the samples modified with TEPA (up to 6 mmol g^{-1}), as shown by Thouchprasitchaia et al. (2018). However, the commercial hydrotalcite tested by Wu et al. (2013) exhibits similar adsorption capacities than the modified LDHs.

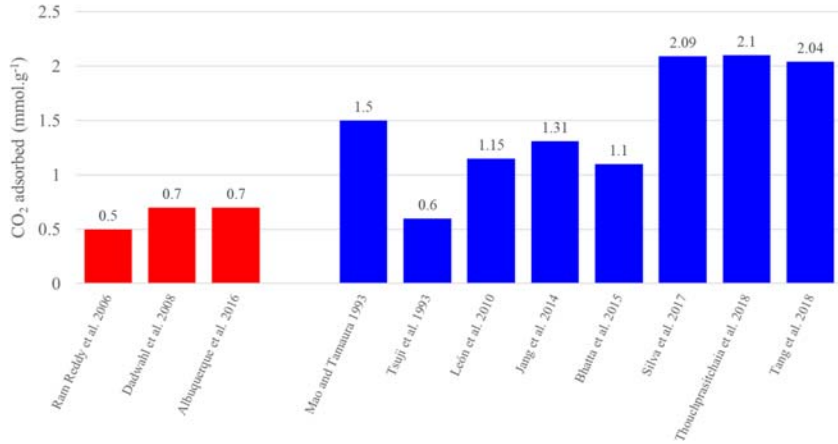
NO_x and SO_x adsorption

Among the wide family of atmospheric pollutants, nitrogen and sulfur-based molecules are emitted from power plants, petroleum refinery, and vehicle engines (Yang et al. 2014). Nitrogen oxides (NO_x) and sulfur oxides (SO_x, a mixture of SO_2 and SO_3) are released into the atmosphere with exhaust gases, causing environmental and climate concerns such as acid rains, photochemical smog, and human diseases (asthma for example) (Corma et al. 1997; Yu et al. 2006; Yang et al. 2014; Wang et al. 2018). The control of these emissions is becoming mandatory due to the increased awareness toward climate change. Moreover, several laws and strict regulations have been put into place in the last decade and encourage

Table 3 CO_2 adsorption by several types of hydrotalcites according to literature

M(II)/M(III) molar ratio	CO_2 adsorbed (mmol g^{-1})	Synthesis process	Modifications/improvements	References
Mg/Al 3	0.5	Co-precipitation	–	Ram Reddy et al. (2006)
Mg/Al 2.87	0.7	Co-precipitation	–	Dadwhal et al. (2008)
Mg/Al 2–3	0.7–1.5	Co-precipitation at pH 10–12.	–	Albuquerque et al. (2016)
Mg/Al 1–5	0.4–1.5	Co-precipitation at pH 10	Fe-promoted	Mao and Tamaura (1993)
Mg, Ni, Zn, Co, Ca, or Cu/Al 1.2	0.15–0.6	Co-precipitation at pH 10	Different M(II) cations	Tsuji et al. (1993)
Mg/Al 3	0.6–1.15	Co-precipitation at pH 10 (low supersaturation)	K- or Na-promoted	León et al. (2010)
Mg/Al 3	1.31	Hydrothermal at pH 11	K-promoted	Jang et al. (2014)
Mg/Al 2	0.42–1.10	Co-precipitation at pH 10–12	K-promoted	Bhatta et al. (2015)
Mg/Al (+Ga) 2	2.09	Co-precipitation	K-promoted + Ga substitution	Silva et al. (2017)
Mg/Al 3	2.1	Co-precipitation	TEPA-supported	Thouchprasitchaia et al. (2018)
Mg/Al 2–3	2.04	Co-precipitation at pH 10	(3-AP)triethoxysilane-grafted	Tang et al. (2018)

Fig. 7 Comparison between nonmodified (red) and modified (blue) hydrotalcites and derived materials on the CO_2 sorption capacity, according to Table 3



research on more efficient ways to reduce them. Several depollution processes have been developed with good results, such as direct decomposition (Yokomichi et al. 1996; Imanaka and Masui 2012; Gunugunuri et al. 2018; Gunugunuri and Roberts 2019), adsorption (Mahzoul et al. 1999; Sedlmair et al. 2003; Atribak et al. 2009), selective catalytic reduction (SCR) (Lukyanov et al. 1995; Forzatti et al. 2010; Gao et al. 2018; Gramigni et al. 2019; Wu et al. 2019b; Li et al. 2019b), and NO_x storage and reduction (NSR) (Castoldi et al. 2006; Breen et al. 2008; Sakano and Kawamura 2018; Umeno et al. 2019). LDHs are interesting due to their efficiency in a wide range of application, as catalysts and adsorbents. In recent years, hydrotalcite-like compounds have exhibited good NO_x and SO_x adsorption capacities. This chapter is dedicated to the adsorption and removal of these atmospheric pollutants by LDHs and related mixed oxides; the synthesis conditions of each LDH are reported in Table 4 with the associated structural modifications.

The most important part of the emitted NO_x comes from the fluid catalytic cracking (FCC) method employed by refinery processes. It consists in converting heavy distillates into lighter ones, such as gasoline and diesel. However, the catalyst gets progressively covered with coke in this process. It then needs to be

Table 4 Synthesis conditions of the LDHs and mixed oxides cited in the part "NO_x and SO_x adsorption"

Reference	Type of material	Surface area ($m^2\ g^{-1}$)	Adsorption capacity ($mmol\ g^{-1}$)	Synthesis method/structural modifications
Corma et al. (1994)	Calcined CuMgAl LDHs	–	–	Co-precipitation at pH 10–13 with fixed Mg/Al molar ratio at 3 and 10 M% Cu, then calcination at 750 °C for 3 h
Palomares et al. (1999)	Calcined CuMgAl and CoMgAl LDHs	117–210	–	Co-precipitation similar to Corma et al. (1994)
Yu et al. (2006)	Calcined CoMgAl LDHs	21.6–162.7	–	Co-precipitation at low supersaturation (pH 10) with change of the Co/Mg molar ratio, then calcination at 800 °C for 4 h
Yu et al. (2007)	Calcined CaCoAl and CaCiLaAl LDHs	–	0.602–0.651	Co-precipitation at low supersaturation (pH 10), then calcination at 800 °C for 4 h
Cheng et al. (2004)	Calcined MgAl LDHs impregnated with Pt	121–249	0.061–0.505	Co-precipitation at low supersaturation (pH 10), then calcination at 600 °C for 5 h and impregnation with $Pt(NH_3)_4(OH)_2$
Silletti et al. (2006)	Pd adsorbed on calcined commercial MgAl LDH	~ 190	0.003–0.062	Commercial LDH with a Mg:Al molar ratio of 7:3 calcined at 600 °C, then $[Pd(acac)_2]$ adsorbed on freshly calcined samples
Li et al. (2007)	Calcined MgRuAl LDHs	280	0.220	Co-precipitation at low supersaturation (pH 10), then calcination at 600 °C for 6 h
Cui et al. (2019)	Calcined MnMgAl LDHs	108–144	0.104–0.426	Co-precipitation at low precipitation (pH 10) assisted by CTAB, then calcination at 600 °C for 6 h
Kameda et al. (2019a, 2019b)	MgAl LDHs	–	–	Co-precipitation at low supersaturation
Cantú et al. (2005)	Calcined MgAl and MgFe LDHs	74–204	–	Co-precipitation at low supersaturation (pH 10)

reactivated by burning and oxidizing the coke in a regenerator unit, leading to the formation of NO_x (Chaparala et al. 2015). The catalysts are reactivated by reduction at 750 °C, so most of the studies are focused on experiments under these conditions. The storage and catalytic reduction of NO_x compounds on the surface of LDH catalysts occurs firstly on the catalytic sites to form nitrites and nitrates, as shown in Fig. 8, and according to the following reactions (Guo et al. 2018):

$$NO + Co^{3+}{-}O^{2-} \rightarrow Co^{2+}{-}(ONO^{-}) \text{ formation of nitrite}$$
$$NO_2 + Co^{3+}{-}O^{2-} \rightarrow Co^{2+}{-}(ONO_2^{-}) \text{ formation of nitrate}$$

Nitrites are unstable on the active sites and are then oxidized to nitrate in the presence of O_2 or decomposed via the following reactions:

$$2NO_2^{-} + O_2 \rightarrow 2NO_3^{-} \text{ by direct oxidation}$$
$$Co^{2+}{-}(ONO^{-}) \rightarrow Co^{3+}{-}O^{2-} + NO \text{ by decomposition}$$
$$2Co^{2+}{-}(ONO^{-}) \rightarrow 2Co^{3+}{-}O^{2-} + N_2 + O_2 \text{ by decomposition}$$

The stored NO_x is finally converted to N_2 and H_2O by reacting with H_2 at 350 °C:

$$2NO^{3-} + 5H_2 \rightarrow N_2 + O^{2-} + 5H_2O$$
$$2NO^{2-} + 3H_2 \rightarrow N_2 + O^{2-} + 3H_2O$$

The first catalytic removal of NO_x using copper-containing LDH-derived mixed oxides has been reported in the late 1990s by Corma et al. (1994). The possibility to incorporate the catalyst in the FCC units was explored because NO_x can be decomposed by the catalyst reduced under a reductive atmosphere at 550 °C, conditions close to those present in the regenerator unit of a FCC plant. The simultaneous removal of NO_x and SO_x in conditions close to those of the regenerator unit was investigated: Cu-mixed oxides derived from hydrotalcite showed to be able to remove both SO_2 (via an oxidation or reduction reaction) and NO (via reduction and/or decomposition) at the same time in the presence of small amounts of O_2 (between 0 and 1.5%). Doping the catalysts with transition metals, such as cobalt, is of great interest on the storage capacity of NO_x in hydrotalcites; Palomares et al. (1999) studied the catalytic activity of CoMgAl mixed oxides derived from hydrotalcites. Besides increasing the surface area through the addition of cobalt, the authors concluded that in the presence of O_2 (up to 1%), the catalyst is able to efficiently remove the atmospheric pollutants. In presence of oxygen in the feed and at 750 °C, cobalt-containing materials showed to be more selective toward NO reduction than Cu-based compounds. Almost 100% of NO conversion was reached in less than 50s and remained stable during the time. Yu et al. (2006) prepared a range of Co_xMg_{3-x}/Al hydrotalcite-like compounds, where x varied from 0 (pure MgAl hydrotalcite) to 3 (pure CoAl hydrotalcite). Despite the decrease of the specific surface area while increasing the cobalt content, higher storage capacities of NO_x were reached at low temperatures, on mixed oxides of intermediate Co concentrations (0.22 mmol g^{-1} at 100 °C for $x = 0.5$). The pure phases exhibited the lowest storage capacities, around 0.14 mmol g^{-1} for MgAl and CoAl hydrotalcites. These results are explained by the increase in the pore size, from 11.6 nm in the pure MgAl mixed oxides to 24.8 nm for the samples with the highest cobalt concentration. Similar experiments were performed the year after by the same team (Yu et al. 2007) with two catalysts: Ca_2CoAl-oxide and $Ca_2CoLa_{0.1}Al_{0.9}$-oxide, derived from their LDH precursors. NO_x storage capacities up to 0.65 mmol g^{-1} at 250 °C were measured. Noble metals such as platinum, palladium, and ruthenium have also been intensively studied toward the storage of NO_x due to their high catalytic redox properties. Cheng et al. (2004) worked on Pt/MgAl oxides: MgAl oxide supports with different molar ratios were obtained by calcination of hydrotalcites, while platinum cations were incorporated by incipient wetness impregnation. The authors concluded that by increasing the Mg/Al molar ratio in the initial hydrotalcites, the stored amount of NO_x at 350 °C was enhanced. Moreover, the addition of platinum as active phase greatly improved the NO_x storage capacity (from 0.06 to 0.50 mmol g^{-1}). NO_x are stored as nitrates that are more stable over Pt/Mg oxides than over Pt/MgAl oxides. A similar approach has been performed by Silletti et al. (2006) with Pd/MgAl oxides. Additional adsorption sites were created, resulting in an increase of 70% in the adsorption capacity, when compared to the MgAl oxides (according to the reaction $PdO_{(s)} + NO_2 \rightarrow Pd(NO_3)_{(s)}$). Finally, RuMgAl oxides have been reported by Li et al. (2007) and exhibited good performances, with a maximum storage capacity of 0.22 mmol g^{-1} at 350 °C.

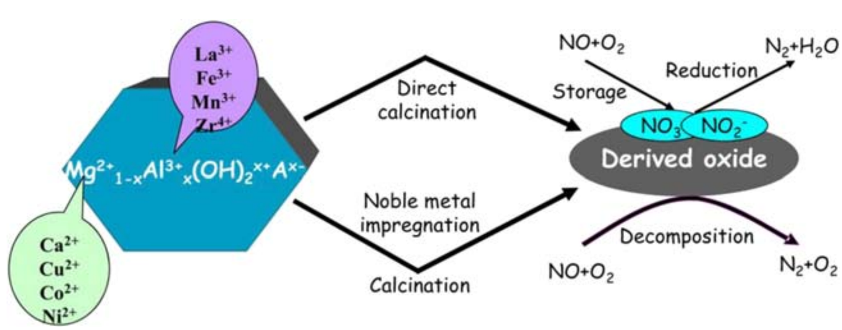


Fig. 8 Schematic representation of LDH-derived catalysts for NO_x storage and reduction (from Yu et al. 2009)

The most recent research focuses on low-temperature storage and release of NO_x between 150 and 300 °C, as reported by Cui et al. (2019) with manganese-doped mixed oxides derived from hydrotalcite precursors. The incorporation of manganese provokes changes in the crystal phases with the formation of Mn_3O_4 and $MgMn_2O_4$ spinel besides brucite. These changes result in the increase of the surface area (from 108 to 139 $m^2\ g^{-1}$) and of the pore size (from 14 to 23 nm), while the presence of manganese favors the oxidation of NO during the adsorption step. The largest adsorption capacity of 0.426 mmol g^{-1} at 300 °C was obtained in the presence of 15 wt% of Mn, as reported in Fig. 9. The efficiency of LDHs containing various metallic elements toward the adsorption of NO_x pollutants has also been reported by Li et al. (2019b) on a novel NiMnTi mixed oxide, showing high surface area and strong reducibility, resulting in strong surface acidity. Moreover, manganese, nickel, and titanium give interesting redox properties to LDHs and gained interest in the last years (Wang et al. 2018; Wu et al. 2019b; Li et al. 2019b). The recent study performed by Kameda et al. (2019a, 2019b) presents another approach on the adsorption of NO_x. The employed catalysts were fresh MgAl hydrotalcites that could be simply recycled by anion desorption in a solution of Na_2CO_3. Low NO removal rate was observed, but the removal of NO_2 was possible and stable at around 50% removal degree, during 3 cycles.

Concerning SO_x removal for industrial applications, several patents are related to the use of hydrotalcites in FCC units (Pinnavaia et al. 1991, 1992; Vierheilig 2003). These compounds are emitted during the oxidative regeneration process of the catalysts at 700–800 °C. The general FCC process applied to SO_x removal is schematized in Fig. 10. Regarding the process conditions, several requirements are needed to trap SO_x (Mathieu et al. 2013): (1) a high thermal stability of the adsorbent under FCC conditions, (2) the ability to oxidize SO_2 to SO_3, (3) the possibility to regenerate the adsorbent under a reductive flow of H_2 to release sulfur as H_2S, (4) the possibility to strongly chemisorb SO_x in the regenerator unit, and (5) the absence of negative impact of the adsorbent on the conversion and selectivity of FCC products. LDHs and hydrotalcites can answer to these requirements: they have the ability to form mixed oxides of the desired elements, stable at high temperatures, and are able to be regenerated in water. The catalytic activity of such compounds strongly depends on the surface area (related to the number of active sites) and on the type of metallic cation employed (for their ability to react with SO_x compounds) (Cantú et al. 2005; Valente and Quintana-Solorzano 2011; Mathieu et al. 2013). The first experiments, focused on MgO, Al_2O_3, and MgAl spinels derived from pure MgAl hydrotalcites, showed very low performances; MgO forms stable $MgSO_4$ compounds and Al_2O_3 forms $Al_2(SO_4)_3$ that is very unstable at these temperatures, leading to the release of sulfate species in the regenerator (Yoo et al. 1992). Three basic reactions are involved in the removal of SO_x in the FCC regenerators (Yoo et al. 1992):

(1) $SO_2 + O_2 \rightarrow SO_3$
(2) $SO_3 + MgO \rightarrow MgSO_4$
(3) $MgSO_4 + 4H_2 \rightarrow MgO + H_2S + 3H_2O$

Based on these equations, several oxides derived from hydrotalcites and LDHs have been tested in the last decade. The most studied samples were prepared by the addition of transition metals similar to those used in NO_x removal processes. Cantú et al. (2005) reported for example on mixed oxides derived from MgAlFeCe hydrotalcites. Despite the lower surface area and low SO_2 adsorption capacity, the oxides containing iron and cerium presented an interesting resistance to deactivation. The addition of cerium increased the adsorption rates and reduced the saturation times, while a limited amount of iron improved the catalytic regeneration of SO_x adsorbents. A similar study was then performed by Valente and Quintana-Solorzano (2011) in partial and full combustion conditions (used for regeneration). The performances of mixed oxides derived from MgAlFeCe

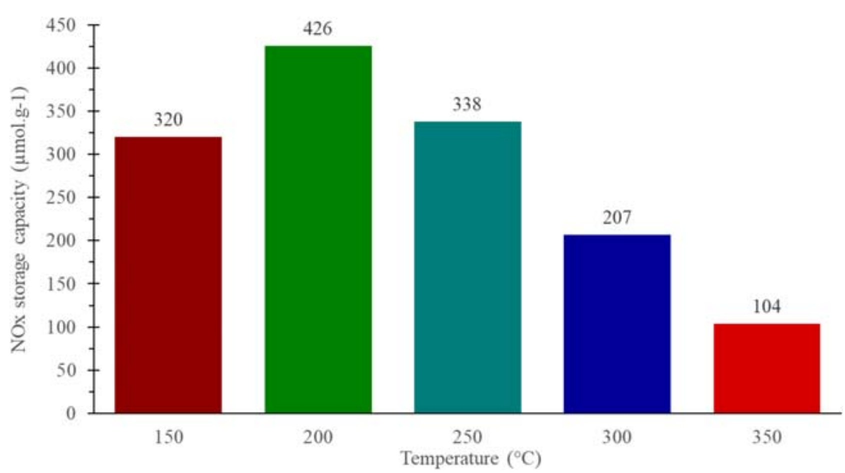


Fig. 9 NO_x storage capacity at different temperatures of mixed oxides derived from MnMgAl LDHs containing 15 wt% Mn (from Cui et al. 2019)

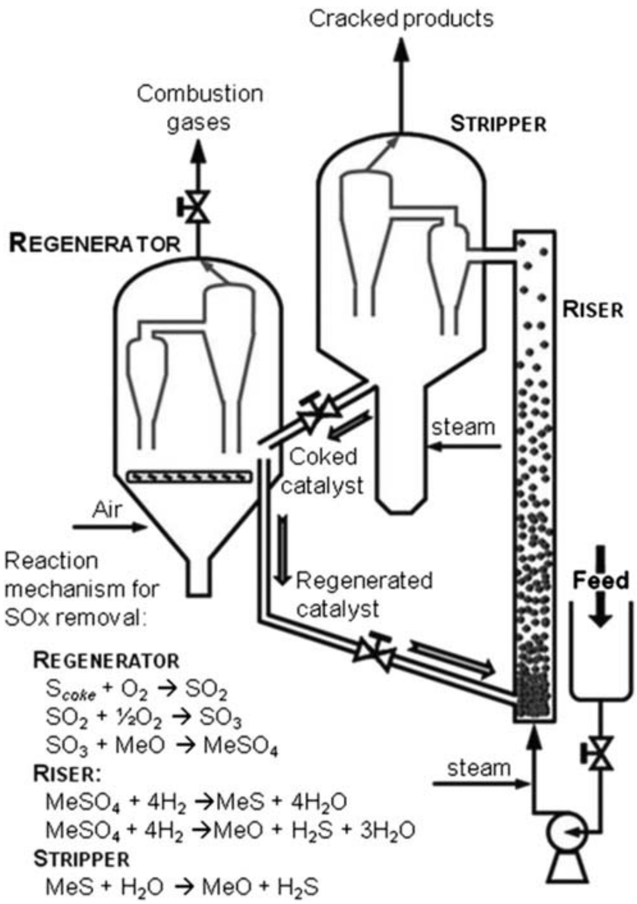


Fig. 10 Simplified scheme of the fluid catalytic cracking (FCC) process adapted to SO_x removal (from Valente and Quintana-Solorzano 2011)

hydrotalcites were compared to those of a commercial catalyst. Hydrotalcites showed a lower deactivation rate and the use of 33% less material than the commercial catalyst. However, only very few recent papers report on the adsorption and removal of SO_x by LDH-based materials. The most recent research focuses on catalysts capable of simultaneously adsorbing SO_x and NO_x (Kameda et al. 2011, 2019a, 2020).

NO_x and SO_x are dangerous compounds formed during the regeneration process of the catalysts employed in the FCC units. Due to their toxicity, their removal has been widely studied. Hydrotalcites and mixed oxides derived from LDHs, in particular those containing transition metals (copper or cerium) or noble metals (platinum, palladium, or ruthenium), have been considered for this application. High specific surface area, large pore sizes, and improved catalytic properties have been obtained by the addition of metallic cations. These modifications led to interesting results in NO_x and SO_x adsorption in regeneration conditions (around 750 °C under a reducing atmosphere). Co-precipitation was the most widely used synthesis method for LDHs, allowing to obtain high specific surface areas and pore sizes and a high number of active sites. Higher adsorption capacities were reached with co-precipitated LDHs (0.10–0.65 mmol g^{-1}), compared to commercial samples (up to 0.06 mmol g^{-1}), as shown in Table 3. The specific surface areas remain in the same range of values (between 100 and 250 m^2 g^{-1}) thanks to the modifications performed on the commercial LDHs. Clearly, the specific surface area is not the only parameter to be taken into consideration for increasing the performances in this application. Some additional studies have been performed on the simultaneous adsorption of NO_x and SO_x in order to obtain a polyvalent compound stable under the conditions of the FCC regenerator units (Kameda et al. 2011, 2019a, 2020).

VOCs total decomposition

Volatile organic compounds (VOCs) give origin to an important class of organic pollutants, such as toluene, ethanol, methanol, acetone, and ethylene (Bahranowski et al. 1999; Carpentier et al. 2002; Gennequin et al. 2010a; Mrad et al. 2015; Kamal et al. 2016). They are emitted from various industrial processes and transport vehicles and represent a serious environmental problem. VOCs are the major contributors to air pollution due to their direct toxic and carcinogenic properties, or indirect contribution to ozone formation. The decomposition of VOCs is thereby crucial for an environmental point of view. VOC removal requires technologies working under hard conditions (temperatures up to 1100 °C and presence of toxic products). The possibility to eliminate VOCs at temperatures lower than those reached during classical combustion is of great interest. Catalytic total combustion of VOCs is then one of the most effective and economically attractive treatment explored in the last decades (Bahranowski et al. 1999; Mikulová et al. 2007; Dula et al. 2007; Tanasoi et al. 2009; Palacio et al. 2010; Kamal et al. 2016). Most catalysts used for the total decomposition of VOCs contain noble metals, such as palladium and gold. Mixed oxides based on transition metals have been also successfully tested with similar activity, as a cheaper alternative. The catalysts must present a high thermal stability and a large number of active sites. Co-precipitation is once again the most adapted method to obtain the desired properties. It permits easily to substitute cations in the structure and to promote the formation of large platelets with a high number of active sites. The synthesis conditions of all the samples cited in this section are listed in Table 5.

Similar to CO_2 adsorption, the total decomposition of VOCs by hydrotalcites and LDHs has been studied on various catalysts presenting different substitutions and in different analytical conditions. The tests were mainly focused on the total decomposition of toluene or ethanol. The first experiments focused on the elimination of VOCs by mixed oxides derived from LDHs have been

Table 5 Synthesis parameters of the LDHs and mixed oxides cited in the section "VOCs total decomposition"

Reference	Type of material	Surface area (m^2 g^{-1})	Synthesis method
Bahranowski et al. (1999)	Calcined CuCr, ZnCr, and CuAl LDHs	CuCr-LDHs 10–20 ZnCr LDH 41 CuAl LDH 56	Co-precipitation at low supersaturation (pH 5.5) and calcination at 600 °C for 3 h. The CuCr-LDHs have been synthesized with M(II):M(III) molar ratios of 1:1, 2:1, and 3:1. ZnCr and CuAl were used to investigate separately the role of Cu and Cr.
Mikulová et al. (2007)	Calcined NiAl LDH	52–161	Co-precipitation at low supersaturation (pH 10) and calcination at 450 °C for 4 h
Gennequin et al. (2009, 2010b)	Calcined CoMgAl LDHs	72–141	Co-precipitation at high supersaturation (pH 9.5) with varying Co:Mg:Al molar ratios of 6:0:2, 4:2:2, 2:4:2, and 0:6:2 and calcination at 500 °C for 4 h
Gennequin et al. (2010a)	Co-deposited calcined MgAl LDHs	3–233	Co-precipitation at high supersaturation (pH 9.5) and calcination at different temperatures (500, 600, 700, 800, and 900 °C). Co was then deposited by aqueous impregnation method.
Palacio et al. (2010)	Calcined MnCuAl and ZnCuAl LDHs	45–108	Co-precipitation at high supersaturation (pH 10) and calcination at 450 and 600 °C
Dula et al. (2007)	Calcined MgMnAl LDHs	137–191	Co-precipitation at low supersaturation (pH 10) and calcination at 600 °C for 4 h. Two different samples have been prepared with Mn either within the structure or in the ionic form as permanganate anions in the interlayer space.
Tanasoi et al. (2009)	Calcined CuMgAl LDHs	114–188	Co-precipitation at low supersaturation (pH 10) with various copper content from 1 to 20 at% and calcination at 750 °C for 8 h
Aguilera et al. (2011)	Calcined MnMgAl, CuMnMgAl, and CoMnMgAl LDHs	75–249	Co-precipitation at low supersaturation (pH 9.5–10.5) with varying Cu/Mn and Co/Mn molar ratios of 0.5, 0.25, 0.1, and 0.05 and calcination at 500 °C for 16 h
Chmielarz et al. (2012)	Calcined CuMgAl, CoMgAl, and CuCoMgAl LDHs	60–217	Co-precipitation at low supersaturation (pH 10) and calcination at 700 or 800 °C. The samples calcined at 800 °C were modified with potassium using aqueous impregnation method and calcined again at 600 °C.
Carpentier et al. (2002)	Calcined MgAl LDH with interlayer Pd complex	33–69	Co-precipitation at low supersaturation (pH 10) in the presence of H_2PdCl_4 and calcination at 290 and 500 °C for 4 h
Kovanda and Jirátová (2011a)	Several calcined LDHs	28–150 (203 with Ni)	Co-precipitation at low supersaturation (pH 10) and calcination at 500 °C for 4 h. NiAl, NiMn, CoAl, CoMn, NiCoAl, NiCoMn, NiCuAl, NiCuMn, CoCuAl, and CoCuMn have been obtained in order to compare with similar mixed oxides deposited on Al_2O_3/Al support.

performed in the late 1990s, even if the reaction was already known for many decades. Moderated temperatures, around 200 °C, are required for this catalytic reaction. The catalysts are mixed oxides and amorphous phases obtained by calcination of LDHs at high temperature. Bahranowski et al. (1999) reported on the impact of Cu(II) and Cr(III) cations in mixed oxides derived from parent LDHs on the total oxidation of toluene and ethanol performed at various temperatures. The authors observed that the purity and crystallinity of the materials depends on the initial Cu:Cr molar ratio and that the catalyst derived from the CuCr hydrotalcite, with an initial Cu:Cr molar ratio of 2, gave the best catalytic performances in both toluene and ethanol oxidations. A similar study, focused on the crystallinity of the LDH precursors, was performed a few years later by Mikulová et al. (2007) on calcined NiAl LDHs. The analyses showed slightly lower catalytic activities than for CuCr hydrotalcite-derived oxides synthesized by Bahranowski et al. (1999) toward toluene oxidation. In ethanol oxidation, the formation of reaction intermediates, such as acetaldehyde, could be avoided by increasing the reaction temperature. Cobalt-containing hydrotalcite precursors with different compositions were tested by Gennequin et al. (2009, 2010b). They showed interesting activities in toluene oxidation. Especially the sample containing the highest amount of cobalt presented the highest specific surface area, well-dispersed oxides, lower presence of cobalt aluminate species, and the highest activity. Figure 11 shows the impact of calcination temperature of Co-containing hydrotalcites on toluene conversion. The same research group also performed studies focused on the memory effect of hydrotalcite (Gennequin et al. 2010a), their ability to regenerate their structure by

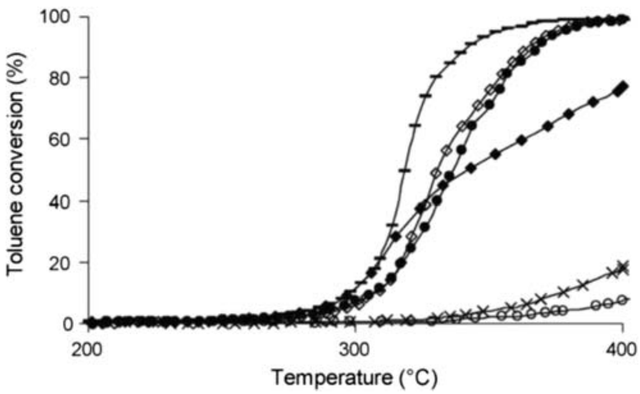


Fig. 11 Toluene conversion versus temperature for Co-containing hydrotalcites previously calcined at various temperatures (from Gennequin et al. 2010a)

simple wetting or impregnation. In this case, cobalt impregnation led to the partial regeneration of the LDH structure and improved toluene oxidation activity.

Palacio et al. (2010) studied the impact of different trimetallic hydrotalcites (ZnCuAl and MnCuAl), after calcination at 450 °C, over the oxidation of toluene. The MnCuAl catalysts showed to be the most efficient (Mrad et al. 2015). Dula et al. (2007) performed a similar study on MgAlMn oxide (obtained from calcination of the parent LDH). The catalyst was more active than MnO_4-impregnated MgAl LDH for the combustion of toluene under air. This behavior can be explained by the enhanced presence of redox sites on the surface of the as-generated mixed oxide particles. Methane is a bit more difficult to oxidize due to its high inertness: its complete oxidation occurs at much higher temperature than toluene and ethanol, as studied by Tanasoi et al. (2009), with 50% of methane decomposition at 480–490 °C over a calcined Cu-containing MgAl-LDH catalyst. This work shows that catalytic activity is dependent on the reducible amount of copper in the catalyst. Aguilera et al. (2011) performed a more complete study on the total oxidation of three different VOCs: butanol, ethanol, and toluene by copper–manganese and cobalt–manganese catalysts derived from the parent LDHs. The catalytic activity depended on the VOC type, and the activity scale was the following for both catalysts: butanol < ethanol < toluene. Some other studies (Chmielarz et al. (2012)) focused on mono-carbon VOCs (methane, methanol, and formic acid). The calcination temperature had a direct impact on complete oxidation: the presence of cobalt phases, obtained at high calcination temperatures (up to 800 °C), induced higher catalytic activities in conversions of methanol and formic acid.

Other modifications, such as the type of interlayer anion or the use of a support, can be introduced in the catalyst conception. Carpentier et al. (2002) demonstrated that the substitution of carbonate anions by palladium complexes in the interlayer space of Mg/Al hydrotalcites improves the catalytic activity. Kovanda and Jirátová (2011a, 2011b) supported several mixed oxides, derived from hydrotalcites, on Al_2O_3/Al or calcined alumina, and it resulted in interesting catalytic activities with the production of side products such as acetaldehyde. Spinel phases identified in the calcined samples are also linked to the high conversion values observed.

Similar to CO_2 adsorption, the total elimination of VOCs remains a very important environmental issue to be solved. Most of the LDHs used in this field are synthesized by the coprecipitation method that makes possible the structural substitution of metallic cations, such as chromium or copper, in order to increase the catalytic activity. Unlike the samples used in NO_x and SO_x adsorption, the materials prepared for VOC elimination, and that containing metal cations in the structure, show similar specific surface area values (between 10 and 250 $m^2\ g^{-1}$) when obtained at low or high supersaturation conditions (Table 5). The main differences in the samples come essentially from the inserted metallic cation and the thermal treatment (temperature and duration). According to the literature, the total elimination of VOCs by LDH-based materials is very promising.

Adsorption of aqueous pollutants

Nowadays, the pollution of water is mainly derived from several industrial effluents (papermaking, textiles, and dyeing for example) and agricultural activities (Chuang et al. 2008; Tzompantzi et al. 2011; Ali 2012). Some of the pollutants released in nature can move into soils and plants. Their high mobility provokes a global concern due to their significant impact on human health, biotoxicity, and carcinogenic effects (Lv et al. 2006; Chuang et al. 2008). As water resource is a fundamental resource on earth and remains rare in some regions, the need to find cheap and efficient depollution processes has increased worldwide. The wide family of aqueous pollutants includes inorganic pollutants (toxic heavy metals such as Pb, Cd, Ni, Zn, and Cr; Lehmann et al. 1999; Yang et al. 2005; El-Sayed et al. 2016; Wan et al. 2017) and organic pollutants (mainly dyes from paper and textile industries; Tzompantzi et al. 2011; Ahmed and Gasser 2012; de Sá et al. 2013; Shan et al. 2014), but also drugs and pharmaceuticals (Chuang et al. 2008; Zhao et al. 2015; Yu et al. 2017)). Many approaches have been proposed for water treatment, including coagulation (Zou et al. 2016a, 2016b), precipitation (Barrera-Díaz et al. 2012; Cheng et al. 2015), sedimentation (Rubí et al. 2009; Guo et al. 2009), sorption (Karickhoff 1984; Fosso-Kankeu et al. 2010; Alila et al. 2011), and ion exchange (Leinonen et al. 1994; Mazur et al. 2016). Regarding the literature, sorption is the most versatile process for removing pollutants at relatively low concentrations, and it is environmentally friendly. This process includes the use of lamellar materials such as clay minerals (Milutinović-Nikolić et al. 2014, p. 99; Wang et al. 2015; Yang et al. 2015) and LDHs

(Yang et al. 2005; Chuang et al. 2008; Ahmed and Gasser 2012; Zou et al. 2017), but also activated carbons (Beita-Sandí et al. 2016; Bedin et al. 2016), graphene (Li et al. 2012; Song et al. 2015; Sun et al. 2017), and metal oxides (Tzompantzi et al. 2011; Cheng et al. 2015; Zou et al. 2017). The main obstacle using these materials is their relatively low adsorption capacity, so the majority of the studies are focused on increasing their efficiency. This part of the review discusses the last advances in the use of LDHs and derived mixed oxides in the adsorption and removal of aqueous pollutants in the last decades. The synthesis parameters of the materials described in this section are listed in Table 6.

Inorganic pollutants are mainly constituted of heavy metals such as arsenic, lead, cadmium, nickel, zinc, and chromium. Their toxicity has been well documented, and their removal gained large focus in the last decades (Barrera-Díaz et al. 2012). Thanks to their accessible and expansible interlayer space, LDHs are interesting adsorbents and ion exchangers of these pollutants. Their use has been firstly documented in the late 1990s and their performances often compared to those of other inorganic sorbents, such as activated carbons and metal oxides (Lehmann et al. 1999). Pure MgAl hydrotalcites generally exhibit interesting removal capacities, up to 50% of Cr(VI) and 100% Zn(II) at pH 4, and are able to act as anion exchangers (in the interlayer space) and proton acceptors (presence of carbonate ions CO_3^{2-}). Moreover, an increase of the solution pH leads to a partial dissolution of the LDH structure. That causes the precipitation of cationic metal species increasing the overall removal of inorganic pollutants. These materials are also efficient toward the removal of trace levels of pollutants, such as arsenic and selenium for example, as described by Yang et al. (2005) by using calcined and uncalcined MgAl hydrotalcites. The adsorption values of uncalcined samples are strongly influenced by the pH of the solution, while no influence of pH was measured on calcined samples (Fig. 12). In both cases, the desorption of As(V) and Se(IV) strongly depended on the anion species, with maximum amounts of 100 ppm in the presence of HPO_4^{2-} and 1000 ppm with SO_4^{2-}, and with desorption yields close to 100%. Several parameters influence the removal capacity by LDHs, such as the pH of the solution, the initial pollutant concentration, and the adsorbent dosage (Lv et al. 2006). Similar to other applications previously discussed, the addition of metallic cations in the structure of LDHs and derived mixed oxides increases the specific surface area and the pore volume of these materials, leading to better sorption

Table 6 Synthesis conditions of the LDHs and mixed oxides cited in the section "Adsorption of aqueous pollutants"

Reference	Type of material	Surface area ($m^2\ g^{-1}$)	Synthesis method
Lehmann et al. (1999)	Commercial MgAl LDH (hydrotalcite)	34.8	Commercial hydrotalcite
Yang et al. (2005)	Calcined MgAl LDHs	47–198	Co-precipitation at high supersaturation, then calcination at 500 °C for 4 h
Lv et al. (2006)	Calcined MgAl, ZnAl, and NiAl LDHs	39.9–240.6	Synthesis method similar to co-precipitation with the use of urea, then calcination at various temperatures (from 200 to 800 °C)
El-Sayed et al. (2016)	Calcined MgAlZn LDHs	65.1	Co-precipitation with 15 wt% Zn, then calcination at 450 °C for 5 h
Xue et al. (2016)	Biochar/MgFe LDHs	3.9 (pores filled by biochar)	LDH directly synthesized in the presence of the biochar by the liquid-phase deposition method (pH 10)
Mahmoud et al. (2017)	Ti–Fe chitosan LDHs	146.5	Ball milling method for 10 h at 300 rpm with titanium isopropoxide, iron nitrate, and chitosan
Yang et al. (2017)	SiO_2 supported on MgAl LDHs	–	In situ co-precipitation method at high supersaturation
Zou et al. (2017)	Calcined CaMgAl LDHs	43–157.8	Hydrothermal synthesis, then calcination at various temperatures (from 200 to 600 °C) for 3 h
Tzompantzi et al. (2011)	Calcined ZnAl LDHs	155–228	Co-precipitation at low supersaturation (pH 9) with different Zn/Al molar ratios, then calcination at 400 °C for 12 h
Seftel et al. (2015)	Various LDHs containing Zn^{2+}, Cu^{2+}, Al^{3+}, Fe^{3+}, or Ti^{4+}	5–230	Co-precipitation at low supersaturation (pH 7.5, 9, or 10) with various M(II)/M(III) molar ratios
Ahmed and Gasser (2012)	MgFe LDHs with Cl^- or CO_3^{2-} as compensating anion	–	Separate nucleation and aging method with colloid mill r otating at 5000 rpm for 3 min
de Sá et al. (2013)	CaAl LDHs	–	Co-precipitation at high supersaturation
Shan et al. (2014)	Magnetic Fe_3O_4/MgAl LDHs	133	Co-precipitation at low supersaturation (pH 10)
Li et al. (2016)	MgAl LDHs	Uncalcined 24.7 Calcined 165.1	Low-temperature ethanol–water-mediated solvothermal method in an autoclave at 190 °C for 1 h

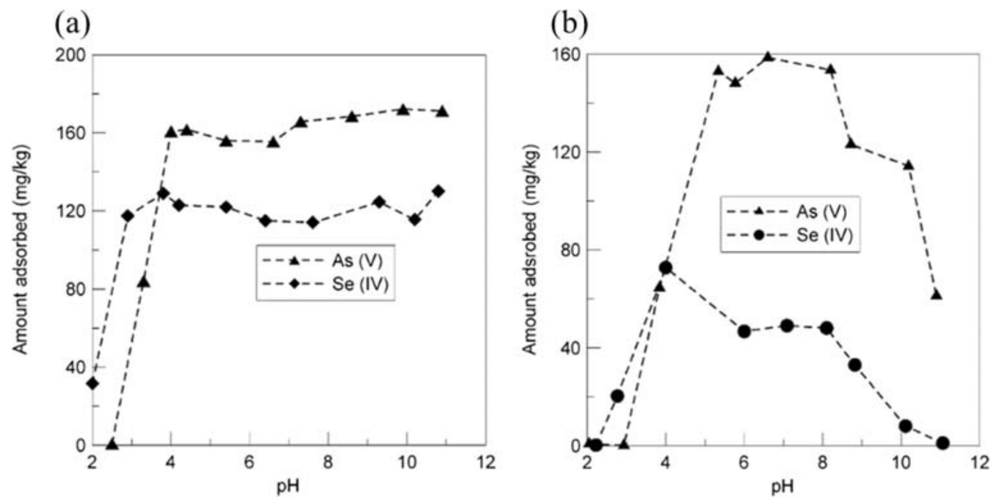


Fig. 12 Effect of pH on the uptake of As(V) and Se(IV) on (**a**) calcined and (**b**) uncalcined LDHs (from Yang et al. 2005)

capacities. El-Sayed et al. (2016) reported the efficiency of MgAlZn mixed oxides derived from calcined hydrotalcites and observed adsorption capacities up to 1.98 and 1.19 mmol $^{-1}$ for Co(II) and Ni(II), respectively. A more recent study showed the positive impact of incorporating LDHs onto a matrix of other nature. Once incorporated into biochar (Xue et al. 2016), MgFe LDHs can uptake nitrates with values up to 0.4 mmol g^{-1}, which are higher than those obtained on MgFe LDHs alone (0.08 mmol g^{-1}) or on activated carbons (0.15 mmol g^{-1}). Chitosan has also been tested due to the possibility to adsorb large amounts of metal ions. Indeed, TiFe LDHs modified with chitosan can exhibit high removal capacities (close to 100%) for cadmium and phosphate in mono- and multiple-pollutant solutions (Mahmoud et al. 2017). The use of hierarchical SiO_2@LDHs on SiO_2 spheres has been studied by Yang et al. (2017) to remove uranium (U(VI)) from aqueous solution. Thanks to the complexations and electrostatic interactions with the abundant oxygen-containing functional groups, a maximum sorption capacity of 1.27 mmol g^{-1} was reported. Similar adsorption capacities have been reported by Zou et al. (2017) using CaMgAl LDHs and derived mixed oxides, with values depending on the calcination temperature of the LDH: from 0.55 (for uncalcined LDH) up to 2.04 mmol g^{-1} (for calcined LDH at 500 °C).

Another family of pollutants regroups the organic aqueous pollutants (pharmaceuticals and molecules with high molecular mass) that are released in nature by chemical industries, creating serious environmental issues due to their high toxicity, even at low concentrations (Chuang et al. 2008; Tzompantzi et al. 2011). The interest in finding a new sorbent for removing such compounds from water increased in the last 10 years, even if the toxicity and the possibility to adsorb these pollutants by natural materials is known since the 1980s (Zepp and Schlotzhauer 1981; Karickhoff 1984). Sorption is one of the most common treatments to remove organic pollutants from nature; LDHs and derived mixed oxides are great candidates thanks to their high sorption capacity in the interlayer region, the large specific surface area and the presence of reactive surfaces (Pavan et al. 1999, 2000; Seki and Yurdakoç 2005; Chuang et al. 2008). Phenolic compounds are the most common organic water pollutants and their adsorption has been widely studied. They can be removed either by simple adsorption or by photocatalytic degradation (Valente et al. 2009b; Paredes et al. 2011; Prince et al. 2015). Both methodologies have been widely studied and reported in the literature. Even if the aim of the present paper is not to review the two mentioned processes, few examples of related applications using LDH are reported as follows. Tzompantzi et al. (2011) reported on the fast photocatalytic degradation of phenols in the presence of ZnAl mixed oxides derived from LDHs. The photodegradation rate is maximized thanks to the electron transfer from Zn^{2+} to Al^{3+}. The removal of phenol molecules by photocatalysis has also been reported by Seftel et al. (2015) using TiO_2 deposited on various LDH matrices presenting different cation compositions (Zn^{2+}, Cu^{2+}, Al^{3+}, Fe^{3+}, and Ti^{4+}). The specific surface area depended on the cations (from 5 m^2 g^{-1} with Cu-substituted LDH to 199 m^2 g^{-1} for Fe-substituted LDH), and almost 90% efficiency toward phenol adsorption by all the TiO_2-modified LDHs was obtained after 5 h of UV and visible light irradiation.

Concerning dye elimination from aqueous solutions, a number of technologies were recently developed, such as adsorption by activated carbon (Özacar and Şengil 2002), nanofiltration (Koyuncu 2002; Lau and Ismail 2009), or photocatalytic degradation (Seftel et al. 2010; Mohapatra and Parida 2012; Yang et al. 2016). However, the formation of sludge in large quantity or the high operation cost makes them not suitable for industrial and large-scale applications (Ahmed and Gasser 2012). The low-cost and regeneration properties of LDHs make the use of such materials interesting, as demonstrated by Ahmed and Gasser (2012) on the removal of Crystal Red, an anionic dye. Figure 13 shows the adsorption

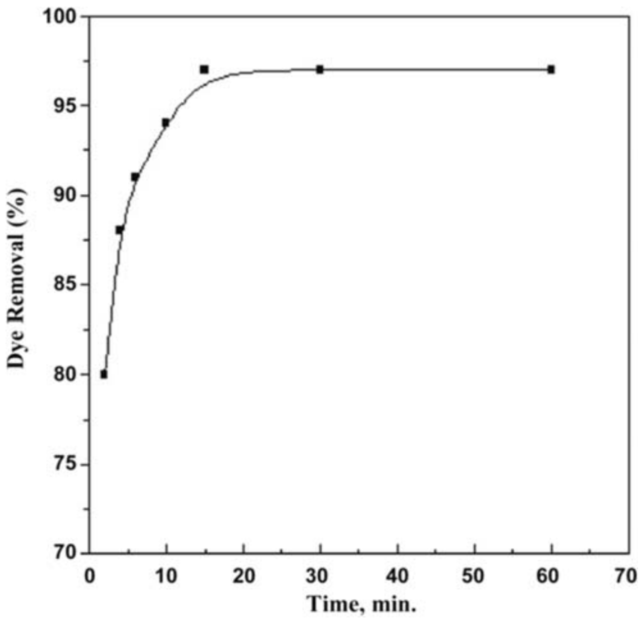


Fig. 13 Effect of contact time on percentage of dye removal Mg–Fe–CO_3–LDH (from Ahmed and Gasser 2012)

of the dye; after only 15 min, 97% of the dye is removed when starting from a solution with a concentration of 100 mg L^{-1}. Moreover, LDH can be regenerated and cycled several times. Similar experiments were also performed by de Sá et al. (2013) with the Sunset Yellow FCF, a synthetic dye used to color food. The increase of the interlayer distance was observed, indicating that the adsorption takes place by simple ion exchange in this area. Magnetic materials can also be great sorbents toward organic dyes, as studied by Shan et al. (2014) by using magnetic Fe_3O_4/MgAl hydrotalcites to remove three different red dyes (reactive red, Congo red, and acid red 1). The uptake equilibrium was reached after 30 min without significant impact on the solution pH. The different results reported all along this review show that the morphology of the LDHs has a great impact on the specific surface area and plays a role on the adsorption capacity. A recent study performed by Li et al. (2016) focused on the adsorption of Congo red by MgAl hydrotalcite. The specific surface area increased from 24.74 to 165.07 m^2 g^{-1} upon calcination of the parent hydrotalcite, due to the enhancement of microporosity derived from the dehydroxylation of the layers and the decomposition of carbonate anions. However, only a slight increase of the adsorption capacity (from 0.186 to 0.206 mmol g^{-1}) was observed. Indeed, the adsorption mechanism implies the anion exchange in the interlayer space and the sorption on the external surface, while the internal surface related to microporosity is not involved in the process due to accessibility problems.

The adsorption of aqueous pollutants, inorganic (toxic heavy metals) and organic (dyes and heavy molecules), by LDHs and derived mixed oxides has been discussed. Due to the increase of industrial and agricultural activities, these dangerous and harmful compounds can be found in high concentrations and need to be removed from aqueous media. Their sorption by lamellar materials, more specifically LDHs, is one of the most investigated solutions. As the adsorption of these pollutants takes place in the interlayer space and on the external surfaces, LDHs and derived mixed oxides with large external specific surface area are required. The co-precipitation method is the most employed because of its ability to form a wide range of LDHs with high crystallinity and reactive surfaces, mainly by substituting magnesium and/or aluminum by other metallic cations in the structure at different molar ratios. According to Table 6, similar specific surface areas were obtained with the samples synthesized by co-precipitation either at low or high supersaturation conditions (between 5 and 230 m^2 g^{-1}). Other synthesis methods such as ball milling (Mahmoud et al. 2017) or the ethanol–water-mediated solvothermal method (Li et al. 2016) are also interesting alternatives to obtain materials with high surface areas (around 150 m^2 g^{-1}). Based on the existing literature, the adsorption of aqueous pollutants on LDHs and their derived materials reveals to be a promising alternative.

Other applications

Hydrogen production

The need of alternative energies to replace fossil fuels is growing since the last decade. Hydrogen is a clean energy vector that can be produced from several processes such as steam reforming, partial oxidation, or SERP (Marquevich et al. 2001; He et al. 2009, 2010; Halabi et al. 2012a, 2012b; Chanburanasiri et al. 2013; Cesar et al. 2016; Homsi et al. 2017). Hydrogen production is carried out in the presence of CO_2 adsorption processes (as reported in the section "CO_2 adsorption—capture and storage"). Sikander et al. (2017) reported in a review article the different hydrogen production processes carried out in the presence of hydrotalcite-base catalysts. The feedstock is generally constituted of light hydrocarbons such as ethanol, methanol, or methane, even if biomass is a promising environment-friendly hydrogen source being highly available worldwide, relatively cheap, and renewable (He et al. 2009, 2010; Contreras et al. 2014; Cesar et al. 2016; Zardin and Perez-Lopez 2017; García-Sancho et al. 2018; Sikander et al. 2018). The hydrogen production reactions are reported in the following equations (de Souza et al. 2012):

(1) $C_2H_5OH + H_2O \rightarrow 2CO + 4H_2$ in the presence of ethanol in water
(2) $C_2H_5OH \rightarrow CH_3CHO + H_2$
(3) $CH_3CHO \rightarrow CH_4 + CO$ by decomposition of ethanol and further decomposition of acetaldehyde

(4) $CH_4 + H_2O \rightarrow CO + 3H_2$ in the presence of methane in water
(5) $CH_4 + CO_2 \rightarrow 2CO + 2H_2$ in the presence of methane in carbon dioxide

The drawback in using hydrotalcite-derived catalysts for these processes is the high operating temperature needed (up to 900 °C) to calcine the layered materials prior to the catalytic reaction. Moreover, the formation of carbon monoxide can lead to the formation of carbonaceous materials (also named coke) that brings to the progressive deactivation of the active sites, as shown by de Souza et al. (2012) in the Boudouard reaction: $2CO \rightarrow CO_2 + C_{(s)}$. The synthesis conditions of the LDHs and related mixed oxides discussed in this section are summarized in Table 7.

The first experiments using LDHs to produce hydrogen were performed, in the last decade, by steam reforming. Marquevich et al. (2001) studied the feasibility with sunflower oil, as feedstock, and Ni/Al mixed oxides derived from calcined LDHs, as catalyst. More recently, steam reforming was studied in the presence of hydrocarbons such as ethanol, methane, or ethylene glycol. For each source, the key point in process development is to identify the most efficient catalysts and synthesize it by varying the synthesis conditions or the chemical composition. Contreras et al. (2014) reported in a review article the various catalysts tested for hydrogen production by steam reforming. Ni-containing catalysts show very good catalytic activity for the conversion of ethanol and ethylene glycol, as shown by He et al. (2009, 2010) and Cesar et al. (2016), respectively. In the first case, the catalytic activity of Co–Ni catalysts decreased by increasing the nickel content. In the presence of ethylene glycol, Cesar et al. (2016) showed that Pt-containing catalysts are the most active and selective. Finally, a complete study on the steam reforming process, carried out with methane as feedstock, has been reported by Halabi et al. (2012a, 2012b), using calcined K_2CO_3-promoted Mg/Al hydrotalcite with in situ CO_2 capture. A similar study has been performed by Chanburanasiri et al. (2013) by comparing the impact of different commercial K_2CO_3 on the sorption efficiency and H_2 production. The co-precipitation method revealed to be the most adapted for the easy substitution of the cations in the structure. In some syntheses, co-precipitation was performed in the presence of urea in order to improve the crystal size and the number of active sites.

Methane decomposition is the second process largely studied for hydrogen production. Methane is firstly adsorbed on the surface of the catalysts and then decomposed on the active sites. Graphite-like structure or coke can accumulate on the active sites and block them, leading to a progressive deactivation of the catalyst. The presence of transition metals (Ni, Co) in the catalyst structure positively contributes to methane decomposition at low temperatures and with high yields. Ashok et al. (2008) identified the ideal composition (Ni/Cu/Al = 60/25/15) to obtain the highest hydrogen yield by varying the Cu/Al molar ratios (maintaining fixed the Ni content) of calcined Ni/Cu/Al hydrotalcites. The amount of Ni in Ni/Mg/Al hydrotalcites can also be modulated; García-Sancho et al. (2017) investigated the influence of nickel content on hydrogen production. Figure 14 shows the methane conversion and the hydrogen production in the presence of hydrotalcites of various nickel contents during the temperature-programmed

Table 7 Synthesis conditions of the LDHs and mixed oxides cited in the section "Hydrogen production"

Reference	Type of material	Surface area ($m^2\ g^{-1}$)	Synthesis method
Marquevich et al. (2001)	Calcined NiAl hydrotalcite	104	Co-precipitation at high supersaturation (pH 9), calcination at 400 °C for 4 h, then reduction under nitrogen steam at 600 °C
He et al. (2009, 2010)	Co-NiMgAl hydrotalcite	Uncalcined 73–144 Calcined 103–180	Co-precipitation at high supersaturation and calcination at 600 °C for 6 h. The metal loading was fixed at 40% and the Co–Ni composition was varied from 40 to 0, 30–10, to 20–20, 10–30, and 0–40.
Cesar et al. (2016)	NiMgAl hydrotalcite	–	Co-precipitation at low supersaturation (pH 13) and calcination at 750 °C for 4 h
Halabi et al. (2012a, 2012b)	Calcined MgAl LDHs	15.6	Commercial hydrotalcite calcined at 400 °C for 4 h and loaded with K_2CO_3 by dry impregnation method
Chanburanasiri et al. (2013)	MgAl LDHs	104	Commercial K_2CO_3-promoted hydrotalcite in cylindrical shape
Ashok et al. (2008)	Calcined NiCuAl LDHs	120–182	Co-precipitation at low supersaturation (pH 9) and calcination at 450 °C for 5 h. Atomic ratio between divalent (Ni^{2+} + Cu^{2+}) and trivalent (Al^{3+}) cations varied from 2 to 9 but constant nickel metal content
García-Sancho et al. (2017)	Calcined NiMgAl LDHs	69–156	Co-precipitation at low supersaturation in the presence of urea and calcination at 850 °C for 10 h
Sikander et al. (2018)	Calcined NiMgAl LDHs	–	Co-precipitation at low supersaturation and calcination at different temperatures of 550, 700, and 800 °C for 6 h

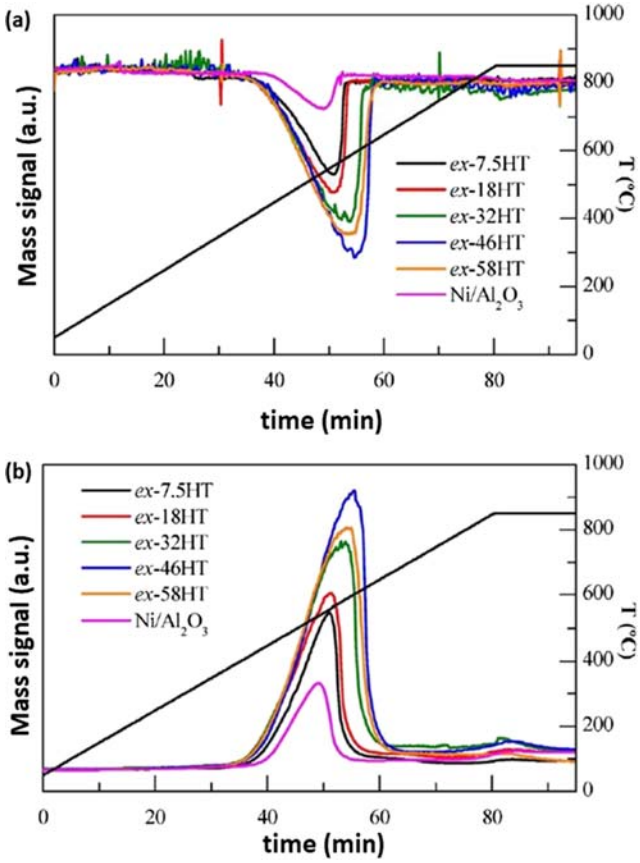


Fig. 14 Temperature-programmed methane conversion (**a**) and hydrogen production (**b**) by nickel-containing calcined hydrotalcites (from García-Sancho et al. 2017)

tests; 55% of methane conversion with 100% hydrogen selectivity was measured on a catalyst containing 46%$_{at}$ of nickel in the structure. The conversion was not improved by increasing the nickel content, due to the presence of inactive isolated Ni particles among the spinel phases. One year later, the same team (García-Sancho et al. 2018) compared the impact of adding nickel during the synthesis of hydrotalcites prior to the calcination and to form bulk Ni-mixed oxides, or after the calcination of the hydrotalcites (to produce Ni-supported mixed oxides). The two kinds of catalysts show similar activities, and only slight differences in the reduction temperatures of the catalysts were observed. The impact of the calcination temperature was studied by Sikander et al. (2018) on Mg/Ni/Al hydrotalcites with different nickel contents. The sample containing 40%$_{at}$ nickel showed the best performances after calcination at 500 and 750 °C. The presence of spinel-like structures on the catalyst surface favors the diffusion of the deposited carbon, enhancing the overall catalytic activity.

Hydrogen can be produced by various processes, such as steam reforming and methane decomposition. Highly active and selective catalysts exist and are listed in Table 8. The possibility to tune the catalyst composition is fundamental to form mixed oxides with high catalytic activity. Most of the studies discussed in this section are focused on the co-precipitation method for the possibility of operating structural substitutions, particularly to add nickel cations in the LDH structure. Table 7 also shows the interest in using co-precipitated LDHs instead of commercial samples; the specific surface areas of the synthesized samples are higher than those of the commercial ones (between 69 and 156 $m^2\ g^{-1}$ instead of 15–104 in the commercial hydrotalcites). Urea is also introduced during the preparation in order to obtain a high number of active sites and increase the size of the platelets. Moreover, hydrogen production can be combined with CO_2

Table 8 H_2 production by several types of hydrotalcite-based materials

M(II)/M(III) molar ratio	Conversion (%)	Selectivity (%)	Synthesis method	Source	References
Ni/Al 2	20–79.7	66.6–72.1	Co-precipitation and calcination at 400 °C for 4 h	Sunflower oil	Marquevich et al. (2001)
Co, Ni, Mg/Al 3	62	70	Co-precipitation and calcination at 600 °C for 6 h	Ethanol	He et al. (2009)
(Zn, Mg, or Co) Ni/Al -	100	70	Co-precipitation, then crushed and sieved to 500–355 μm and calcined at 600 °C for 6 h	Ethanol	de Souza et al. (2012)
Mg/Al 1.56	72–99.6	91.3–99.8	Commercial Htlc calcined at 400 °C for 4 h	Methane	Halabi et al. (2012b)
(Ni, Mg)/Al 1	49–94	56.1–96.5	Co-precipitation and calcination at 750 °C for 4 h	Ethylene glycol	Cesar et al. (2016)
M(II)/M(III) molar ratio	Conversion (%)	H_2 produced (a.u.)	Synthesis method	Source	References
(Ni, Mg)/Al 3	15–55	246–1253	Urea + co-precipitation and calcination at 850 °C for 10 h	Methane	García-Sancho et al. (2017)
(Ni, Mg)/Al 3	15–40	250–1100	Urea + co-precipitation and calcination at 850 °C for 10 h	Methane	García-Sancho et al. (2018)

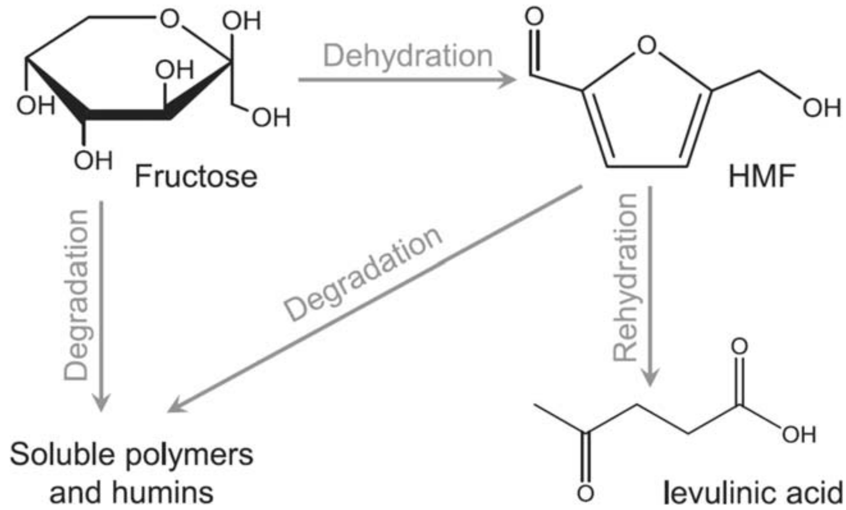


Fig. 15 Scheme of the reactions involved in fructose dehydration and the resulting formation of 5-HMF and by-products (from Dou et al. 2018)

adsorption with interesting results. The combination of CO_2 capture and H_2 formation in the same process is promising for future industrial applications.

5-Hydroxymethylfurfural formation

Thanks to its natural abundance, biomass is a promising sustainable feedstock: carbohydrates are the largest natural carbon source constituting up to 75% of the annual biomass production, with promising applications in chemistry, food, paper, and pharmaceutical industries (Rosatella et al. 2011; Zakrzewska et al. 2011). Carbohydrates are rich in oxygen and can be easily reduced and dehydrated into a large spectrum of chemicals such as 5-hydroxymethylfurfural (5-HMF) or levulinic acid. The US Department of Energy listed 5-HMF as among the top 10 value-added chemicals; it is generally used as an intermediate to synthesize a wide range of chemicals nowadays derived from petroleum. Unfortunately, 5-HMF is not extensively produced at an industrial scale due to the high production cost (van Putten et al. 2013). Fructose and glucose conversion are well-documented reactions that occur at relatively low temperature (below 150 °C). These mild reaction conditions allow the use of clays, hybrids, and other lamellar materials as catalysts. The dehydration of fructose leads to the formation of 5-HMF, the main reaction product, and other by-products, such as levulinic acid and formic acid that are formed by degradation of 5-HMF, as explained in Fig. 15. The catalysts used in this reaction must present a high specific surface area in order to maximize the catalytic surface and react with the fructose dissolved in the solvent (generally water). These requirements can be fulfilled by LDHs synthesized by the sol–gel method or any other method implying a hydrothermal treatment, and that present high specific surface areas and an improved thermal stability. The synthesis parameters of all the cited samples are reported in Table 9.

Table 9 Synthesis parameters of the lamellar materials cited in the section "5-Hydroxymethylfurfural formation"

Reference	Type of material	Surface area ($m^2\ g^{-1}$)	Synthesis method
Lourvanij and Rorrer (1994)	Pillared Na-montmorillonite	137.7–249.6	Na-montmorillonite mixed into the pillaring agent solution
Fang et al. (2014)	Cr-montmorillonite K10	–	K-10 montmorillonite exchanged with Cr cations
Chheda and Dumesic (2007)	Calcined MgAl LDH, MgO/ZrO_2, and MgO/TiO_2	~300	MgAl LDH synthesized by co-precipitation at low supersaturation and mixed oxides by the sol–gel method. All these samples have been calcined at 450 °C for 8 h.
Li et al. (2011)	Calcined NiMgAl LDH	34–117	Co-precipitation at low supersaturation (pH 10) with varying Ni at%, and calcination at 800 °C for 5 h
Zeng et al. (2009b)	Calcined AlZr LDHs	66–281	Co-precipitation at high supersaturation (pH 9) and calcination at 500 °C for 6 h

The main drawback in using clays, hybrids, and other lamellar materials in catalytic reactions is their thermal stability. In typical natural clays, dehydration occurs below 200 °C with the removal of interlayer and surface water, while irreversible dehydroxylation with formation of oxides starts already around 350 °C, depending on the type of metallic cations in the starting material. Several catalytic reactions occur above 300 °C; thus, the majority of the studies performed with LDHs are based on mixed oxide catalysts obtained by their calcination. However, few experiments have been performed on

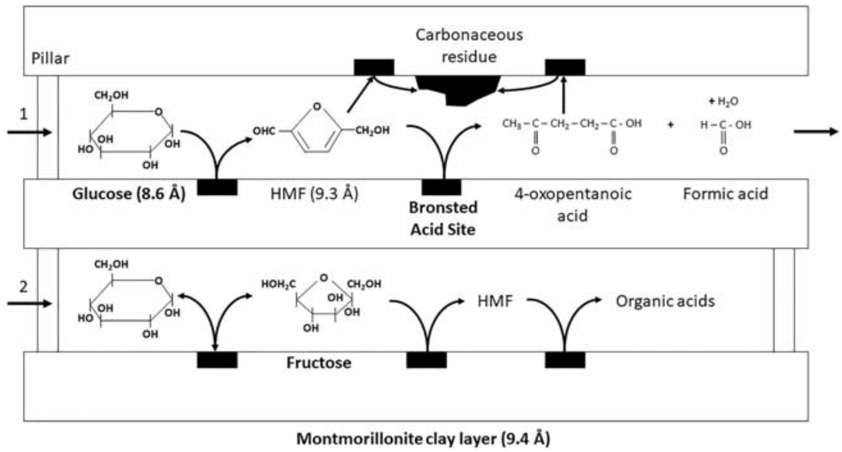


Fig. 16 Proposed intraparticle diffusion and reaction scheme for glucose and fructose within a pillared clay catalyst. Reaction 1: sequential dehydration of glucose to HMF and organic acid. Reaction 2: isomerization of glucose to fructose (from Lourvanij and Rorrer 1994)

modified clay minerals such as montmorillonite; thanks to their expandable layers and swelling properties, these minerals can be easily modified by ion exchange and pillaring in order to modify their structural properties. Lourvanij and Rorrer (1994) studied the partial dehydration of glucose to organic acids on iron, chromium, aluminum, and nonpillared montmorillonite. Their large pore size (higher than 10 Å) allows the diffusion of glucose molecules (equal to 8.6 Å) in the interlayer space, as demonstrated by Fig. 16. Depending on the reaction path, coke formation, leading to their progressive deactivation, has been observed. Further studies have been performed by Fang et al. (2014) with a commercial montmorillonite (K-10) to compare the impact of the reaction solvent: dimethyl sulfoxide (DMSO) and butyl-3-methylimidazole chloride ([BMIM]Cl) were used. The results are shown in Fig. 17 and point out the higher conversion of glucose in the presence of higher concentrations of [BMIM]Cl in DMSO. The efficient recycling of the solvent (up to six times) without significant loss of the catalytic activity was also observed.

Besides the well-known basicity of hydrotalcites, LDHs, and related materials, only few studies on the dehydration of fructose in 5-hydroxymethylfurfural have been performed on these materials. However, recent research pointed out the promising application of these materials in the transformation of biomass-derived monomers (Chheda and Dumesic 2007; Li et al. 2011; Climent et al. 2014). Zeolites are widely employed for such applications, but mixed oxides represent a great alternative thanks to their easy synthesis by the sol–gel method or by simple thermal treatment of LDHs and hydrotalcites.

In the case of glucose catalysis, three different reactions can be expected: dehydration, isomerization, and retro-aldol condensation. The first reaction generally allows to form HMF and levulinic acid in the presence of acid catalysts. On the other hand, the other reactions are favored in the presence of basic materials to form fructose or lactic acid. The bifunctional acid–base properties of Al–Zr mixed oxides bring to different reaction products: HMF and levulinic acid are formed by dehydration on the acid sites, while lactic acid and fructose are obtained by isomerization and retro-aldol condensation in the basic sites.

As explained in this section, the catalytic formation of 5-HMF by dehydration of glucose with the use of hydrotalcites or LDHs is not much documented, but the potential use of their related mixed oxides shows interesting results: they are promising for future research as zeolite substitutes and needs further investigations. Even if the specific surface areas of the LDHs reported in Table 9 are similar to those of modified montmorillonites, the catalytic activity in this application is related to the surface acidity, limiting the potential use of

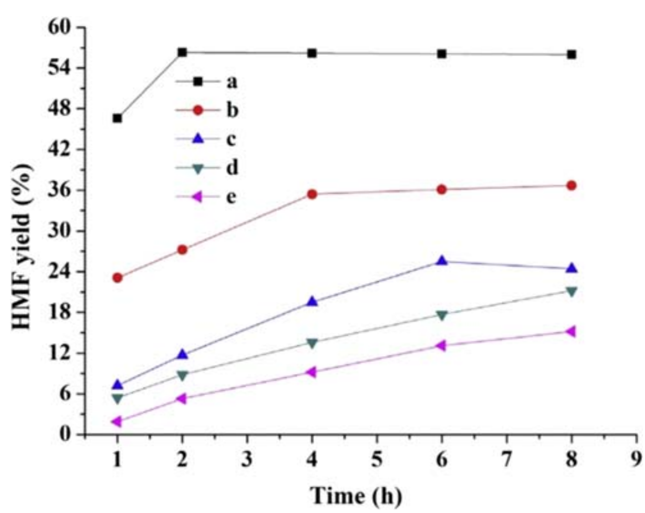


Fig. 17 Catalytic conversion of glucose into 5-HMF at various [BMIM]Cl concentrations in DMSO, e = DMSO only (from Fang et al. 2014)

LDHs. Moreover, the sol–gel method or hydrothermal treatment can be easily performed to obtain LDHs adapted to this application.

Conclusion

LDHs, hydrotalcites, and their related materials gained huge interest in the last decades thanks to their simple synthesis processes and numerous applications. Even if they are mainly synthesized by co-precipitation or hydrothermal methods, the possibility to combine different M(II) and M(III) cations and interlayer anions in their structure allows to tailor the chemical composition, as well as their structural properties (surface areas, pore sizes, number of active sites), making them potential catalysts in several industrial and environmental processes. Moreover, their memory effect allows to reuse these catalytic materials several times, which is interesting at industrial scales.

In this review, the impact of synthesis methods on the structural and textural properties of LDHs has been discussed. The aim was to identify the most suitable synthesis process for a chosen environmental application. The adsorption of pollutants (atmospheric and aqueous) and hydrogen production are important environmental applications in which LDHs can find their place. The key parameter for implementing the use of LDHs and derived materials is the choice of the appropriate synthesis process. The adsorption of pollutants requires a high amount of basic sites and strong thermal stability that can be obtained using the co-precipitation method to synthesize LDHs. The formation of 5-HMF takes place on the acid sites that can be potentially obtained by sol–gel or hydrothermal synthesis. Finally, the intercalation, impregnation, or addition of appropriate elements (metals) with uniform dispersion is important for hydrogen production to increase the catalytic activity of hydrotalcites.

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Abbreviations *5-HMF,* 5-hydroxymethylfurfural; *DMSO,* dimethyl sulfoxide; *HPLC,* high-performance liquid chromatography; *LDH,* layered double hydroxide; *NMR,* nuclear magnetic resonance; *SERP,* sorption-enhanced reaction process; *TEPA,* teraethylenepentamine; *VOCs,* volatile organic compounds

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